nature portfolio

Peer Review File

Interferon stimulated gene GALNT2 restricts respiratory virus infections

Corresponding Author: Professor Jincun Zhao

Version 0:

Decision Letter:

10th February 2025

*Please ensure you delete the link to your author homepage in this e-mail if you wish to forward it to your co-authors.

Dear Professor Zhao,

Thank you for your patience while your manuscript "A novel interferon stimulated gene GALNT2 is required to restrict respiratory virus infections" was under peer-review at Nature Microbiology. It has now been seen by 3 referees, whose expertise and comments you will find at the end of this email. Although they find your work of some potential interest, they have raised a number of concerns that will need to be addressed before we can consider publication of the work in Nature Microbiology.

We consider that your revision plan is sensible and if further experimental data allow you to address the reviewers' criticisms, we would be happy to look at a revised manuscript.

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We strongly support public availability of data. Please place the data used in your paper into a public data repository, if one exists, or alternatively, present the data as Source Data or Supplementary Information. If data can only be shared on request, please explain why in your Data Availability Statement, and also in the correspondence with your editor. For some data types, deposition in a public repository is mandatory - more information on our data deposition policies and available repositories can be found at https://www.nature.com/nature-research/editorial-policies/reporting-standards#availability-of-data.

Please include a data availability statement as a separate section after Methods but before references, under the heading "Data Availability". This section should inform readers about the availability of the data used to support the conclusions of your study. This information includes accession codes to public repositories (data banks for protein, DNA or RNA sequences, microarray, proteomics data etc…), references to source data published alongside the paper, unique identifiers such as URLs to data repository entries, or data set DOIs, and any other statement about data availability. At a minimum, you should include the following statement: “The data that support the findings of this study are available from the corresponding author upon request”, mentioning any restrictions on availability. If DOIs are provided, we also strongly encourage including these in the Reference list (authors, title, publisher (repository name), identifier, year). For more guidance on how to write this section please see: http://www.nature.com/authors/policies/data/data-availability-statements-data-citations.pdf

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If you wish to submit a suitably revised manuscript we would hope to receive it within 6 months. If you cannot send it within this time, please let us know. We will be happy to consider your revision, even if a similar study has been accepted for publication at Nature Microbiology or published elsewhere (up to a maximum of 6 months).

In the meantime we hope that you find our referees' comments helpful.

Yours sincerely,
[Signature redacted]

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Reviewer Expertise:

Referee #1: Viral restriction factors
Referee #2: Coronavirus
Referee #3: Glycobiology

Reviewer Comments:

Reviewer #1 (Remarks to the Author):

Ran and colleagues examined the ability of GALNT2, a novel interferon-stimulated gene (ISG), in restricting the replication of respiratory viruses, including SARS-CoV-2 and influenza A virus (IAV). The authors provide evidence that GALNT2 is upregulated in response to interferon signaling in mice and humans and impairs proteolytic processing and fusogenic activity of viral glycoproteins by modulating O-linked glycosylation. They further show that mutations in specific serine residues (S810/813) of the SARS-CoV-2 spike protein affect sensitivity to GALNT2-mediated restriction. Finally, they provide evidence that GALNT2 loss-of-function variants in humans increase the risk of hospitalization due to SARS-CoV-2 infection.

Altogether, the manuscript provides strong evidence that GALNT2 exerts inhibitory effects in at least two respiratory viral infections. The data are of significant interest but several points need to be addressed.

1. It is interesting that individuals with GALNT2 loss-of-function variants seem to have an elevated risk of hospitalization due to SARS-CoV-2 infection. If feasible, the authors should clarify if these individuals may experience any other defects or health issues beyond virus susceptibility.

2. The authors should briefly discuss why GALNT2's role as an antiviral ISG may have been missed in previous studies and screens for antiviral factors.

3. The patient data in Figure 1g is based on a very small cohort, and the analysis is somewhat unclear. The authors should elaborate on how the data were processed and why GALNT2 levels appears to be upregulated in one condition (pseudo-bulk)

and downregulated in another (ciliated cells).

4. The authors suggest that the main impact of IFN is GALNT2-dependent. However, the results do not fully support this given that the STAT knockout (KO) shows a similar effect as the GALNT2 KO. This should be addressed by providing more detailed analysis or cautioning the interpretation of the data.

5. In Figure 4, the authors show that the higher band of the spike protein, often associated with fully processed and infectious spike, actually represents the O-glycosylated inhibitory form. Is it possible to separate the functional spike from the GALNT2-modified spike? Where does the spike protein appear in the absence of GALNT2?

6. The authors should briefly explain why they consider plaque size (readout in Figure 5) as the best measure of viral inhibition. Also, the results are difficult to comprehend. It seems that in this case, the absence of O-linked glycosylation reduces viral replication since the results show a significant reduction in viral titers and plaque size for both recombinant viruses (S810/813I and S810/813A) compared to the WT SARS-CoV-2. This suggests that O-linked glycosylation benefits the virus, possibly through TMPRSS2-dependent activation. However, while the absence of O-linked glycosylation seems to reduce viral replication, the explanation of why this occurs is not fully clear. The authors suggest that the O-linked glycosylation benefits viral replication, yet the data show reduction in viral titers. This contradiction should be addressed more explicitly, potentially considering whether the spike mutations at S810/813 have a role beyond GALNT2 restriction, including a potential effect on TMPRSS2-dependent activation.

7. What are GALNT2 expression levels normalized on in Fig. 1e and 1f? Panel 1h is unclear and labels too small. Why is the 72 hpi time point not shown for Omicron in Fig. 2a? Fig. 5e: how can there be an mRNA change of GALNT2 in infected GALNT2 -/- mice?

8. Do WT and IFNAR-/- mice express similar levels of ACE2 after Ad5-hACE2 transduction?

9. Some figure legends should provide more experimental details, such as the number of replicates, independent repeats, and the specific meaning of error bars. The English is faulty at some place.

Reviewer #2 (Remarks to the Author):

This manuscript descibes the antiviral activity of O-GalNAc transferase 2 (GALNT2) as interferon regulated gene with antiviral activity against SARS-CoV-2 and influenza A virus. GALNT2 is regulated by type-I IFN both in vitro and in vivo and its depletion and overexpression in vitro and in an AAV overexpression model in vivo, leads to increased or decreased virus growth and pathology, respectively. Following these findings, the authors used a series of in vitro experiments, viral like particles and recombinant SARS-CoV-2 virus to identify S810/813 as one GALNT2 glycosylation site. They found that infection of viruses and VLPs generated in GALNT2 expressing cells is impaired, which is corroborated by cell-cell fusion assays showing reduced fusion of cells in the presence of GALTN2 and spike of SARS-CoV-2. Most interesting is the identification of a glycosylation site at positions S810/813, mutations rendered viruses insensitive to GALNT2, supporting that the identified sites are relevant for virus growth. Overall the authors propose a model whereby viruses released from cells expressing GALNT2 would be inactive for entering neighboring cells. The authors propose a mechanism of virus inactivation – in response to IFN or more specifically GALNT2 – which would limit virus growth and disease severity.

This is an interesting manuscript describing GALNT2 as antiviral protein and providing a potential mechanism for its activity. The data is supported by in vitro, in cell and in vivo evidence. Most intriguing is the notion that GALNT2 expressing cells would release inactive viruses, which are incapable to infect cells, but which may even serve as activators of immune cells in infected tissue. The study thus adds a conceptual advancement on the activity of the innate immune system. My overall enthusiasm is slightly affected by insufficient descriptions of experimental procedures, missing raw or processed data, data not being deployed in public repositories or as supplementary data, which affected reviewing these datasets. However, I support publication of this manuscript since it is an interesting concept and may be an important aspect of antiviral immunity. Additional points need to be addressed, though.

Major points
1. Methods, analysed data and all source data on the presented RNAseq, LC-MS/MS and glycoproteomics analyses and associated statistical analyses are missing in material and methods or supporting material. Primary data in online repositories is missing.
2. Statistics - the authors should use SD instead of SEM, please indicate the number and nature of datapoints that were considered.
3. Figure 1: The authors use an ACE2 expressing AAV vector to transduce IFNAR KO mice with ACE2 and thereby show that a type-I dependent mechanism is restricting SARS-CoV-2 in vivo. I understand that expression of ACE2 in IFNAR KO mice is only a tool to increase infectability and to lead to the identification of GALNT2. However, an alternative interpretation of this experiment could be that the ACE2 overexpressing AAV vector is better infecting IFNAR KO mice (e.g. more ACE2 expression or different distribution) and the described differential infectability in IFNAR KO vs wt mice is driven by differential expression patterns of ACE2. I feel that the authors need to clarify this point. Best would be immunohistochemistry analysis of ACE2 expression in both mouse strains. An additional option would be to plot the ACE2 expression levels in the RNAseq dataset of Fig 1a (which is unfortunately not provided within the current version of the manuscript).
4. Figure 1c: The authors are transiently transfecting pcDNA3 based plasmids into Huh7 cells and infect these cells with SARS-CoV-2 for N staining. I am not a big fan of such assays since heterogeneity of the transfection efficacy of the individual plasmids and the differential strength of overexpression of individual proteins can lead to heterogenous protein expression, differentially

affect cell viability and thereby general viral propagation. Thus this experiment may be limited in its interpretability. Moreover, overexpression of individual proteins (such as IFIH1) may change the overall antiviral state of the cells without much specificity to the given infection. This concern may be reflected by the fact that that all 28 tested ISGs are somewhat antiviral… it is simply uncommon that all transcriptionally regulated genes have antiviral activity. I suggest to use stably transduced cells for Figure 1c with control for gene expression or to remove these data.
5. This paper generally employs diverse means of protein overexpression using plasmid transfection, stable cell lines, lentiviral transduction. It is often unclear from figure legends which type of overexpression has been used. This is an important variable in the given experiments, please clarify this for every experiment shown. For some transient plasmid-transfection mediated overexpression experiments it would be important to show transfection efficacy (% of infected cells) since untransfected cells would influence some of the assays.
6. The authors describe GALNT2 as yet unstudied ISG. Therefore additional information and relation to canonically used ISGs would be very interesting to readers. The manuscript may benefit from additional information on tissue distribution and its already known regulation in diseases – such data can easily be provided by consulting public databases.
7. Figure 1e/f shows the dynamics of GALNT2 – how does this compare to canonical ISGs such as OASs, IFITs or STATs or USP18, SOCS (which are negative regulators expressed after the first wave of ISGs)? To me the expression of GALNT2 appears to to rise quite late in wt cells. Also, please indicate which method was used for quantification and show the raw data (western blots?) – this is not described. Similarly, it would be nice to plot the dynamics of expression data of known ISGs in the scRNAseq experiment of Fig 1i/j (e.g. as extended data). I acknowledge that the authors profile wt and GALNT2 KO cells using transcriptomics, proteomics and glycoproteomics. I feel that characterization of the same cells in infected conditions could be informative for readers to understand the function of GALNT2 in infected conditions.
8. Figure 1d, Figure 2: Please indicate how the overexpression cells were generated – are these stable cells? Does GALNT2 expression affect cells overall? The DAPI staining in Figure 1d looks more dull – does this reflect reduced cell gowth? It is important to characterize the used cells by testing the cellular metabolism (e.g. ATP levels by Cell titre glow or similar) and evaluate the growth kinetics of these cells over time. It is also not clear from the figure legends how statistics were determined. These are representative experiments of repeats and the small error bars indicate that these are based on some technical (how many?) replicates. Statistics and average values considering all biological repeats would be reassuring to evaluate the effect.
9. Figure 2c (and extended Figure 3): An essential control of transfecting an unrelated protein is missing to exclude cellular resistance induced by the transfection procedure. It would be further essential to know transient transfection efficacy (of GALNT2 and currently missing control) in these cells.
10. I am a bit surprised by the expression of GALTN2 in THP-I cells by IFN-lamda treatment. THP-I cells appear not to express IFN-lambda receptor (also evidenced by the reduced STAT1 levels) – what is inducing GALTN2 in these cells? Can the authors please plot for an alternative ISG in this cell type?
11. Figure 1f: figure legend mentions the expression of STAT1 in IFNAR KO mice, but these blots are not shown. It would be an interesting control, though.
12. Extended Figure 5e, g – it is unclear to me what “fold change” means in this case… which conditions were compared? It would maybe be better to plot delta-CT values vs a housekeeping gene.
13. Extended Figure 5f –the authors mention it in the text that GALNT2 depletion affected lung pathology and show one example of wt and GALNT2 KOs – it would be nice if the authors could this across the different mice tested. I understand that the shown lungs are very supportive there.
14. Extended Figure 6c- I (and other in vivo data): Currently a variable number between 3 and 6 datapoints are plotted (though often coming from one experiment). Please plot not detected values (you can indicate detection limits accordingly). It would further be nice to indicate the number of mice that were assessed in these experiments in Figure legends.
15. Figure 3d, e; Extended Figure 6g-k: Can the authors exclude that GALNT2 expression affects subsequent infection with AAV-ACE2? ACE2 is required for SARS-CoV-2 infection and impaired expression may affect this in vivo model. For the interpretability of the data, potential impairment of AAV-ACE by GALNT2 expression would be important to be evaluated.
16. Extended Figure 8a: The authors mention that the plaque size of viruses derived from GALTN2 overexpressing cells is smaller. I do not understand why this is the case – would the cells used for plaque assay not promote expression of viruses? I.e. if a virus infects the cells, should it not be propagated similarly, independent of the virus-producing cell in the cell monolayer?
17. It is very interesting that influenza HA is inhibited by GALNT2. Does this also apply to HA serotypes with a polybasic cleavage site, such as HAs derived from avian influenza viruses?
18. The identification of the glycosylation site at positions S810/813 is very interesting and well done. Particularly growth assays in presence of GALNT2 overexpression are supporting this story. Since GALNT2 has been reported to be quite promiscuous in activity, it would be very reassuring if the glycosylation assay shown in figure 4g was repeated with the S810/813 mutant(s) viruses. What is the occupancy of glycosylation on virus particles that are released from wt and GALTN2 KO and IFNAR or STAT1 KO cells? One would expect that this glycosylation pattern is prominent in wt cells but not in cells lacking GALNT2 or IFN signaling.

Reviewer #3 (Remarks to the Author):

In “A novel interferon stimulated gene GALNT2 is required to restrict respiratory virus infections” Ran et al. set out to explore interferon stimulated genes in the pulmonary tissue of IFNAR KO mice infected with SARS-CoV2.
The authors identify several genes that are differentially downregulated in the WT and IFNAR KO background after SARS-CoV-2 infection. They then overexpress all 28 genes in Huh7 cells and quantified percentage of infected cells after SARS-CoV-2 infections. Here GALNT2 stands out as the one reducing infection the most.
Now focusing only on GALNT2, GALNT2 transcript levels are analyzed in single-cell RNA-seq data from BALF and PBMCs from healthy controls and COVID-19 patients – mild and severe cases. The authors state that high GALNT2 transcript level is associated with COVID-19 patients compared to Healthy Controls, but more with mild than severe disease.
Based on a number of cell-based in vitro assays and animal KO/overexpression of GALNT2 the authors conclude that GALNT2

is an interferon responsive gene with antiviral activity towards SRARS-CoV-2 as well as Influenza A virus.
The authors explore the impact of GALNT2 overexpression and conclude that there is no major impact on the host cell by measures of transcriptomics/proteomics and glycoproteomics and speculate that GALNT2 functions through targeting viral proteins. Using in vitro enzymatic glycosyltransferase activity assays, they then test the activity of GALNT2 on peptides from the spike protein covering a furin and a TMPRSS2 cleavage site. This data leads the authors to test if GALNT2 activity changes processing of recombinantly expressed spike and HA with similar cleavage sites. The authors conclude that both Spike and HA are differentially processed after GALNT2 activity and suggest that impaired viral entry could be part of the anti-viral mechanism. In cell-cell fusion assays etc.

Comments

In general, the figures are laid out in a logic manner and the manuscript flow is relatively easy to follow. The manuscript text could benefit from some sort of language service/proofreading.

Query 1. One argument the authors provide that GALNT2 is anti-viral relates to four alleged GALNT2 LoF variants identified in the GnomAD database. Prior studies exploring such variants present in the broad population suggested none or only minor effect on the enzymatic activity (e.g. Guda et al., PNAS 2009, Hansen et al., Glycobiology 2015). Revisiting this database, this reviewer did not find any of the suggested variants to be predicted to be deleterious and two are even validated as benign. Maybe the authors tested the enzymatic activity of the 4 variants (V554M, V552M, D314A, Q216H)? If so, please provide this data to support the conclusion that GALNT2 activity is anti-viral.

Query 2. Several GALNT2 LoF animal models have been generated and published, and significantly reduced body weight as well as a short stature phenotype have consistently been reported in Galnt2 deficient mice, as well as in humans and dairy cattle (Edmondson et al., BioRxiv 2024, Mercanoglu et al., Sci Rep 2024, Yang et al., PNAS 2023, Verzijl et al., Mol Metab 2022, Zilmer et al., Brain 2020, Khetarpal et al., Cell Metab 2016, Reynolds et al., Nat Genet 2021). Thus, the data presented in Fig. 3 a where WT and Galnt2 KO animals have same body weight is a bit confusing. And this reviewer wonders if the reduced body weight observed post infection is an inherent phenotype of the animal. If the authors have generated a LoF model that is different from previously published, they should note this and present growth data. In general, the authors should discuss their findings in relation to the established literature.

Query 3. To explore the impact of Galnt2 overexpression the authors take a multi-omics approach with transcriptomics/proteomics/glycoproteomics and identify significant changes at all three levels, but do not discuss or add value to these changes. Interestingly Galnt2 overexpression in mouse pancreas is lethal at the age of 4-6 weeks and other studies have demonstrated dose-dependent activities of GALNT2 (Mercanoglu et al., Sci Rep 2024, Hintze et al., JBC 2019). Of note, several of the upregulated proteomics/glycoprotomics hits are ER residents, which could be suggestive of a stressed secretory pathway, as often seen when driving high protein expression. Maybe there are technical details worth exploring in this context, but this reviewer failed to find the experimental section describing transcriptomics/proteomics/glycoproteomics experiments? How were these experiments conducted?

Query 4. GALNT2 is part of a large family of isoenzymes with largely redundant activities. To substantiate their idea that GALNT2 specifically has anti-viral activity, the authors express several GALNT isoenzymes to test in vitro activity with spike and HA peptides. The authors do not present data to suggest that the other isoforms are active but write that a peptide from muc1 is tested against all isoforms demonstrating enzymatic activity. The authors should provide this data as at least one of the isoforms should not work on this peptide? (Coelho et al., JACS 2022)

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Version 1:

Decision Letter:

29th August 2025

*Please ensure you delete the link to your author homepage in this e-mail if you wish to forward it to your co-authors.

Dear Professor Zhao,

Thank you for your patience while your manuscript "A novel interferon stimulated gene GALNT2 is required to restrict respiratory virus infections" was under peer-review at Nature Microbiology. It has now been seen by 3 referees, whose expertise and comments you will find at the of this email. You will see from their comments below that while they find your work of interest, some important points are raised. We are very interested in the possibility of publishing your study in Nature Microbiology, but would like to consider your response to these concerns in the form of a revised manuscript before we make a final decision on publication.

In particular, you will see that reviewer #3 still raises some important issues that will need to be addressed. The rest referees' reports are clear and the remaining issues should be straightforward to address.

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Figure legends must provide a brief description of the figure and the symbols used, within 350 words, including definitions of any error bars employed in the figures.

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EXTENDED DATA FIGURES

When re-submitting your manuscript, please ensure that any supplementary figures and tables that are crucial to the manuscript's conclusions are converted into Extended Data figures and tables to increase visibility of these data. Extended Data figures and tables are online-only (present in the online PDF and full-text HTML versions of the paper), peer-reviewed display items that provide essential background to the article but are not included in the main article due to space constraints. A maximum of ten Extended Data display items (figures and tables) is permitted.

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We hope to receive your revised paper within three weeks. If you cannot send it within this time, please let us know.

We look forward to hearing from you soon.

Yours sincerely,
[Signature redacted]

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Reviewers Comments:

Reviewer #1 (Remarks to the Author):

The authors have addressed my concerns and I congratulate them to an interesting study.

Reviewer #2 (Remarks to the Author):

I want to thank the authors for thoroughly addresssing my points.
This is an excellent manuscript, a pleaure to read - I fully support its publicaiton.

Note: Fig 4l: Hoescht33342 should be Hoechst33342

Reviewer #3 (Remarks to the Author):

For Query#1
The MALDI-TOF time course experiment is not quantitative, and the conclusion written in the manuscript text (Line 334-338) is exaggerated and incorrect. Based on this experiment the authors cannot conclude that the activity of GALNT2 is markedly decreased in the missense variants and should adjust their conclusions accordingly.

Line 213 - Variants found in the GnomAD NOT GenomeAD

The authors use of LOF in relation to an enzyme (here GALNT2) should be defined as conventionally this refers to loss of activity and is misleading in the context presented in the abstract (line 60) and in the manuscript (line 212.

Query#2
No further comments

Query#3
GALNT2 is NOT localized in the ER, but in the Golgi.
And the conclusion line 301 seems exaggerated and incorrect. The findings of ER proteins should be mentioned and may be a consequence of cellular stress.

Query#4
No further comments

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Version 2:

Decision Letter:

16th September 2025

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Dear Professor Zhao,

Thank you for your patience while your manuscript "A novel interferon stimulated gene GALNT2 is required to restrict respiratory virus infections" was under peer review at Nature Microbiology. It has now been seen by our referees, and in the light of their advice I am delighted to say that we can in principle offer to publish it. First, however, we would like you to revise your paper to address the points made by the reviewers, and to ensure that it is in Nature Microbiology format.

The referees' remaining comments are clear, and should not be difficult to implement. Editorially, we will need you to make some changes so that the paper complies with our Guide to Authors at http://www.nature.com/nmicrobiol/info/gta.

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In recognition of the time and expertise our reviewers provide to Nature Microbiology's editorial process, we would like to formally acknowledge their contribution to the external peer review of your manuscript entitled "A novel interferon stimulated gene GALNT2 is required to restrict respiratory virus infections". For those reviewers who give their assent, we will be publishing their names alongside the published article.

Specific points:
In particular, while checking through the manuscript and associated files, we noticed the following specific points which we will need you to address:

[LIST ANY POINTS ARISING FROM THE AIP CHECKLIST HERE]

General points:

We will also need you to check through all of the following general points when preparing the final version of your manuscript:
We estimate that your manuscript currently exceeds our normal length limit for Articles of about 3,000 words. We have some flexibility, and can allow a revised manuscript at 3,500 words, but please consider this a firm upper limit. You could achieve some shortening by moving some details to the Methods section that should follow the main text (the length of the Methods section is unlimited and does not count towards the main text length).

Titles should give an idea of the main finding of the paper and ideally not exceed 90 characters (including spaces). We discourage the use of active verbs and do not allow punctuation.

The paper's summary paragraph (about 150-200 words; no references) should serve both as a general introduction to the topic, and as a brief, non-technical summary of your main results and their implications. It should start by outlining the background to your work (why the topic is important) and the main question you have addressed (the specific problem that initiated your research), before going on to describe your new observations, main conclusions and their general implications. Because we hope that scientists across the wider microbiology community will be interested in your work, the first paragraph should be as accessible as possible, explaining essential but specialised terms concisely. We suggest you show your summary paragraph to colleagues in other fields to uncover any problematic concepts.

Please include a data availability statement as a separate section after Methods but before references, under the heading "Data Availability". This section should inform readers about the availability of the data used to support the conclusions of your study. This information includes accession codes to public repositories (data banks for protein, DNA or RNA sequences, microarray, proteomics data etc…), references to source data published alongside the paper, unique identifiers such as URLs to data repository entries, or data set DOIs, and any other statement about data availability. At a minimum, you should include the following statement: "The data that support the findings of this study are available from the corresponding author upon request", mentioning any restrictions on availability. If DOIs are provided, we also strongly encourage including these in the Reference list (authors, title, publisher (repository name), identifier, year). For more guidance on how to write this section please see:
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Please check the PDF of the whole paper and figures (on our manuscript tracking system) VERY CAREFULLY when you submit the revised manuscript. This will be used as the 'reference copy' to make sure no details (such as Greek letters or symbols) have gone missing during file-transfer/conversion and re-drawing.

All Supplementary Information must be submitted in accordance with the instructions in the attached Inventory of Supporting Information, and should fit into one of three categories:

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We will not send your revised paper for further review if, in the editors' judgement, the referees' comments on the present version have been addressed. If the revised paper is in Nature Microbiology format, in accessible style and of appropriate length, we shall accept it for publication immediately.

Please resubmit electronically

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* publication-quality figures. For more details, please refer to our Figure Guidelines, which is available here: https://mts-nmicrobiol.nature.com/letters/Figure_guidelines.pdf

* Extended Data & Supplementary Information, as instructed

* a point-by-point response to any issues raised by our referees and to any editorial suggestions.

* any suggestions for cover illustrations, which should be provided at high resolution as electronic files. Please note that such pictures should be selected more for their aesthetic appeal than for their scientific content. I am sure you will understand that we cannot make any promise as to whether any of your suggestions might be selected for the cover of Nature Microbiology.

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We hope that you will support this initiative and supply the required information. Should you have any query or comments, please do not hesitate to contact me.

We hope to hear from you within two weeks; please let us know if the revision process is likely to take longer.

Yours sincerely,

[Signature redacted]

****
Reviewer Expertise:

Referee #1:
Referee #2:
Referee #3:

Reviewer Comments:

****

Version 3:

Decision Letter:

22nd October 2025

Dear Professor Zhao,

I am pleased to accept your Article "Interferon stimulated gene GALNT2 restricts respiratory virus infections" for publication in Nature Microbiology. Thank you for having chosen to submit your work to us and many congratulations.

Over the next few weeks, your paper will be copyedited to ensure that it conforms to Nature Microbiology style. We look particularly carefully at the titles of all papers to ensure that they are relatively brief and understandable.

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Responses to reviewers' comments

Reviewer #1 (Remarks to the Author):

Ran and colleagues examined the ability of GALNT2, a novel interferon-stimulated gene (ISG), in restricting the replication of respiratory viruses, including SARS-CoV-2 and influenza A virus (IAV). The authors provide evidence that GALNT2 is upregulated in response to interferon signaling in mice and humans and impairs proteolytic processing and fusogenic activity of viral glycoproteins by modulating O-linked glycosylation. They further show that mutations in specific serine residues (S810/813) of the SARS-CoV-2 spike protein affect sensitivity to GALNT2-mediated restriction. Finally, they provide evidence that GALNT2 loss-of-function variants in humans increase the risk of hospitalization due to SARS-CoV-2 infection.

Altogether, the manuscript provides strong evidence that GALNT2 exerts inhibitory effects in at least two respiratory viral infections. The data are of significant interest but several points need to be addressed.

We sincerely appreciate the constructive feedback and positive evaluation of our work provided by the reviewer.

1. It is interesting that individuals with GALNT2 loss-of-function variants seem to have an elevated risk of hospitalization due to SARS-CoV-2 infection. If feasible, the authors should clarify if these individuals may experience any other defects or health issues beyond virus susceptibility.

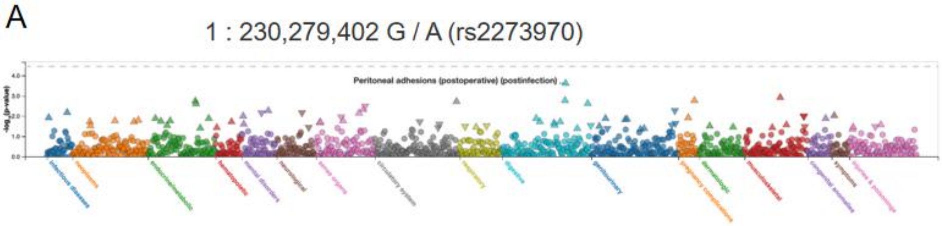
We thank the reviewer for this insightful comment. To assess whether GALNT2 loss-of-function (LoF) variants are associated with broader health effects beyond viral susceptibility, we analyzed the UK Biobank TOPMed-imputed PheWAS dataset (https://pheweb.org/UKB-TOPMed/), which includes approximately 1,400 curated health phenotypes and 57 million imputed variants in ~400,000 White British individuals.

We first evaluated the association between the GALNT2 LoF variant identified in our study (rs2273970) and the 1,400 phenotypes. No associations reached suggestive significance ($P < 10^{-5}$), indicating that rs2273970 is not strongly linked to other major disease conditions in this large population cohort (Figure A).

To further investigate whether genetic variants regulating GALNT2 expression are associated with broader health phenotypes, we performed additional PheWAS analyses using two top cis-eQTLs for GALNT2, rs750556 and rs750557, which were consistently identified across multiple immune cell types in the DICE database (https://dice-database.org/genes/GALNT2). Again, no significant associations were observed at a suggestive threshold ($P < 10^{-5}$) for either variant (Figure B).

Together, these analyses suggest that GALNT2 LoF and expression-regulating variants do not appear to be associated with other major disease traits in the general population.

A



B

GALNT2 eQTL

SNP ID	Cell type	Position (chr1)	Padj(eQTL)	Effect size
rs750557	B cell, naive	229253206	0.0046	-0.74
rs750556	B cell, naive	229253129	0.022	-0.77

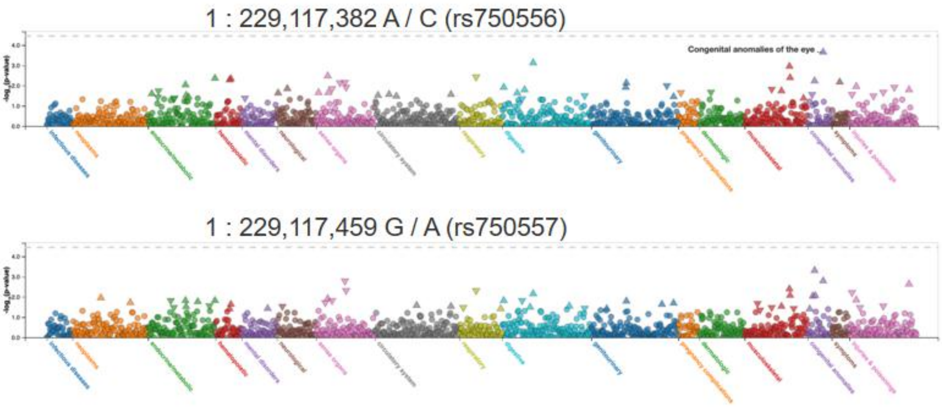


Figure legend: Phenome-wide association analysis (PheWAS) of GALNT2-related variants using UK Biobank TOPMed-imputed data. (A) PheWAS of the GALNT2 loss-of-function variant rs2273970 across approximately 1,400 curated phenotypes from ~400,000 White British individuals in the UK Biobank. Each dot represents the association of rs2273970 with a specific phenotype; the y-axis shows the $-\log_{10}$ (P value). Phenotypes are grouped by category on the x-axis. The horizontal dashed line indicates the suggestive significance threshold ($P = 10^{-5}$). (B) PheWAS of two cis-eQTLs of GALNT2, rs750556 and rs750557, using the same dataset. Each panel shows the associations between one eQTL and the full set of phenotypes. No traits reached the suggestive significance threshold ($P = 10^{-5}$). The top panel summarizes the eQTLs' basic information from the DICE database. All analyses were conducted using the publicly available PheWeb platform (https://pheweb.org/UKB-TOPMed/).

2. The authors should briefly discuss why GALNT2's role as an antiviral ISG may have been missed in previous studies and screens for antiviral factors.

Different interferon-stimulated genes (ISGs) induced by different viral infections are differentially regulated and they also exhibit varying levels of background expression across different cell lines (*in vitro* screening), leading to divergent outcomes in various screening models (PMID: 21478870, PMID: 34581622, PMID: 37380775). This variability often results in researchers focusing primarily on top candidate genes, thereby potentially overlooking GALNT2. Our screening study was based on in vivo experiments and integrated sequencing data from clinical samples (Fig. 1 and extended Data Fig. 1-2) for a comprehensive analysis that led us to identify multiple candidates. Several well-known ISGs were identified, suggesting the robustness of our screening system. Furthermore, we focused on the novel candidate, GALNT2, to

explore the undefined antiviral strategy. Through this whole study, we finally added a missing piece of the puzzle - that O-glycosylation can be directly regulated by innate immunity - highlighting the diversity of antiviral strategies employed by human immunity.
We have incorporated this point into the discussion in the revised manuscript (lines 486-494).

3. The patient data in Figure 1g is based on a very small cohort, and the analysis is somewhat unclear. The authors should elaborate on how the data were processed and why GALNT2 levels appears to be upregulated in one condition (pseudo-bulk) and downregulated in another (ciliated cells).
We thank the reviewer for this valuable comment. In our study, we integrated publicly available single-cell RNA-seq data from bronchoalveolar lavage fluid (BALF) samples across 9 independent studies, comprising 59 severe COVID-19 patients, 5 mild patients, and 28 healthy controls (HCs). We performed both pseudo-bulk and cell-type-specific analyses to comprehensively assess GALNT2 expression changes.

In the pseudo-bulk analysis, gene expression was averaged across all cell types within each patient, allowing us to assess global expression differences between disease groups. This analysis revealed that GALNT2 levels were elevated in both mild and severe patients compared to healthy controls, with no significant difference between the mild and severe groups. This pattern reflects a general upregulation of GALNT2 across multiple cell types, including ciliated, secretory, myeloid, T and B cells (Fig. 1c, Extended Data Fig. 2d).

In contrast, our cell-type-specific analysis revealed a more nuanced pattern: GALNT2 expression was significantly higher in ciliated cells from mild patients compared to those from severe patients, while no such difference was observed in other cell types. As ciliated cells are among the primary targets of SARS-CoV-2 infection, the elevated GALNT2 expression in this specific cell type in mild cases may suggest a protective or regulatory role in mitigating disease severity.

The apparent discrepancy between the pseudo-bulk and single-cell results arises from the averaging nature of pseudo-bulk analysis, which can obscure cell-type-specific changes. We now clarify this in the revised manuscript.

To improve clarity, we have made the following revisions:
In the Fig. 1c legend, we now explicitly state that the top panel shows pseudo-bulk expression and the bottom panel shows ciliated cell-specific expression, and explain the rationale for comparing both.

In the Methods section (lines 747-749), we now describe how pseudo-bulk expression was calculated (i.e., summing raw counts per gene across all cells per patient,

followed by normalization and log-transformation) and how cell-type annotations were used to extract cell-type-specific expression profiles.
We hope these changes better clarify the analytical approach and interpretation of the data.

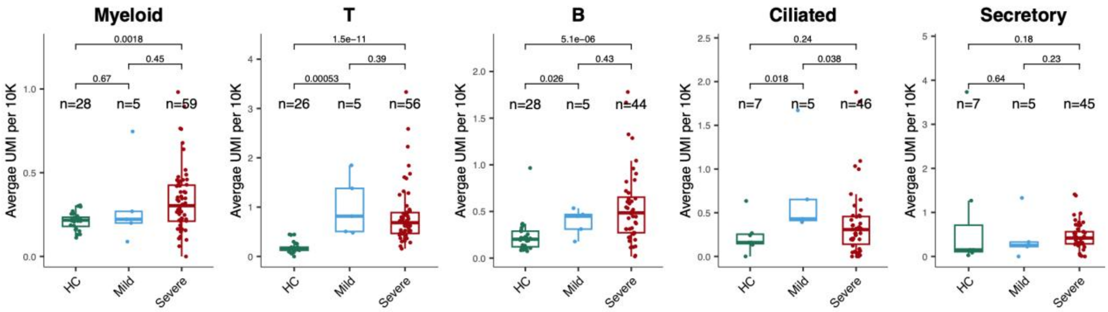


Figure legend: Comparison of GALNT2 expression in Myeloid, T, B, ciliated and Secretory cells of BALF samples using scRNA-seq. Each dot represents an individual, and the y-axis represents the average expression level of GALNT2 across all cells or specifically in ciliated cells from that individual. The p values were calculated by Wilcoxon signed-rank test.

4. The authors suggest that the main impact of IFN is GALNT2-dependent. However, the results do not fully support this given that the STAT knockout (KO) shows a similar effect as the GALNT2 KO. This should be addressed by providing more detailed analysis or cautioning the interpretation of the data.
Interferons (IFNs) induced JAK-STAT signaling pathways contribute to innate immune recognition and an anti-viral state (as shown below). We have already proven that GALNT2 is a critical effector regulated by the IFNs and STAT1 signaling pathway and plays a significant role in mediating IFN's antiviral effects in our study. We found that GALNT2 is essential for restricting viral infections, as evidenced by the following observations:
(1) STAT1 knockout (STAT1$^{-/-}$) cells exhibit a similar phenotype to GALNT2 knockout (GALNT2 KO) cells (Fig. 2c).
(2) Treatment with IFN-β in Huh7, Huh7 GALNT2$^{-/-}$, and Huh7 STAT1$^{-/-}$ cells followed by PR8 infection demonstrated that GALNT2 functions as a crucial effector, contributing significantly to the inhibition of viral infection (Extended Data Fig. 6f).
(3) Overexpression of GALNT2 in STAT1$^{-/-}$ cells and mice markedly restores antiviral activity, further indicating that GALNT2 is indispensable for restricting viral infections (Fig. 2c, 3e).
Based on our data, we acknowledged that GALNT2 is only one of the most critical ISGs against SARS-CoV-2 infection. We are cautious when interpreting our data and provided further discussion in our revised manuscript (lines 236-238).

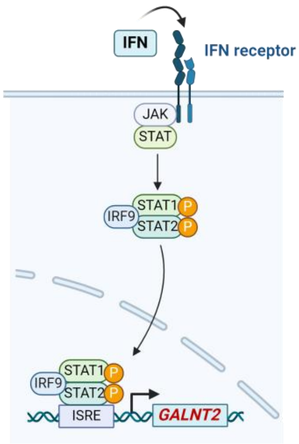


Figure legend: Schematic model to show GALNT2 is a novel ISG. Canonical type I IFN signaling activates the Janus kinase (JAK)-signal transducer and activator of transcription (STAT) pathway, leading to transcription of ISGs, including GALNT2.

5. In Figure 4, the authors show that the higher band of the spike protein, often associated with fully processed and infectious spike, actually represents the O-glycosylated inhibitory form. Is it possible to separate the functional spike from the GALNT2-modified spike? Where does the spike protein appear in the absence of GALNT2?
The molecular weight difference between the functional spike and the GALNT2-modified spike is minimal. In the absence of GALNT2, molecular weight of full-length spike is ~190kDa and the cleaved S2 is around ~98kDa. To distinguish the GALNT2-modified spike from the total spike protein, we employed an HPA pulldown assay using biotin-conjugated Helix promatia agglutinin (biotin-HPA), a lectin that specifically detects *O*-linked N-acetylgalactosamine (*O*-GalNAc) modification with streptavidin magnetic beads (Fig 4f). The results indicate that the molecular weights are similar (Fig. 4h: spike in HPA pulldown vs. input group), making it very difficult to differentiate the two forms of the protein using SDS-PAGE.

6. The authors should briefly explain why they consider plaque size (readout in Figure 5) as the best measure of viral inhibition. Also, the results are difficult to comprehend. It seems that in this case, the absence of O-linked glycosylation reduces viral replication since the results show a significant reduction in viral titers and plaque size for both recombinant viruses (S810/813I and S810/813A) compared to the WT SARS-CoV-2. This suggests that O-linked glycosylation benefits the virus, possibly through TMPRSS2-dependent activation. However, while the absence of O-linked glycosylation seems to reduce viral replication, the explanation of why this occurs is not fully clear. The authors suggest that the O-linked glycosylation benefits viral replication, yet the data show reduction in viral titers. This contradiction should be addressed more explicitly, potentially considering whether the spike mutations at S810/813 have a role beyond GALNT2 restriction, including a potential effect on TMPRSS2-dependent activation.
We thank the reviewer for these comments.

(1) The formation of a viral plaque is an important phenomenon during influenza A virus and SARS-CoV-2 infections. Plaque size is commonly employed to assess viral infectivity and fusion capability.

(2) The two serine residues 810/813 of SARS-CoV-2 spike are essential for TMPRSS2 recognition and cleavage, as we previously reported (PMID: 38744813). In agreement with this observation, these two mutated viruses exhibit smaller plaque size and reduced replication and pathogenicity both *in vitro* and *in vivo* due to diminished utilization of host TMPRSS2 for viral entry. On the other hand, although the substitutions of S810/813 with either Isoleucine (I) or Alanine (A) prevent O-glycosylation and enable viral evasion of GALNT2-dependent antiviral activity (Fig. 5f and 5g), they also impaired TMPRSS2-dependent viral-cell fusion and virus and virus replication. Therefore, our results indicate that GALNT2-mediated innate immunity exerts a highly conserved antiviral effect, making it difficult for the virus to evade this defense mechanism through mutation. It also explained why there was only a transient occurrence of I813 substitution in B.1.36.25 towards the end of 2020, which quickly disappeared (Fig. 5a).

We have discussed the results in the manuscript in more depth (lines 463-470 and 541-556).

7. What are GALNT2 expression levels normalized on in Fig. 1e and 1f? Panel 1h is unclear and labels too small. Why is the 72 hpi time point not shown for Omicron in Fig. 2a? Fig. 5e: how can there be an mRNA change of GALNT2 in infected GALNT2 -/- mice?

We appreciate the reviewer's valuable feedback.

(1) We included Western blot (WB) results below to provide additional support for Fig. 1e-f (Extended Data Fig.1f and 1g in the revised manuscript) (as shown below). The protein levels of GALNT2 in the lungs of WT and IFNAR$^{-/-}$ mice were quantified by grayscale and subsequently subjected to statistical analysis. We included these data in the Extended Data Fig.1e in our manuscript.

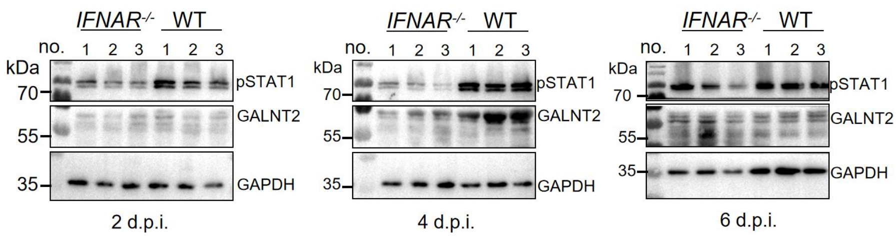


Figure legend: Western blot analysis of mouse GALNT2 protein expression in SARS-CoV-2-infected WT and *IFNAR*$^{-/-}$ mouse lungs at 2, 4, 6 d.p.i. after Ad5-hACE2 transduction.

(2) We enlarged Fig. 1h (Fig. 1b, in the revised manuscript) to enhance its clarity.

(3) We added data on viral titers at 72 hours post-infection (h.p.i.) following Omicron infection in the revised manuscript (Fig. 2a).

(4) It is important to note that we did not observe any changes in GALNT2 mRNA levels in infected GALNT2$^{-/-}$ mice upon viral infection. Here we showed the fold change compared with GAPDH by qPCR assay with high sensitivity or some extent of

background readings. In fact, we confirmed the GALNT2 knockout in mice by intranasally stimulation with 2 μg of IFN-α2 and were unable to detect GALNT2 expression via Western blot (Extended Data Fig. 7c). (After intranasally administration of IFN-α2 for 2, 4, or 6 days, GALNT2 knockout in mice was confirmed. Compared to wild-type (WT) mice, no upregulation of GALNT2 protein was detected in Galnt2 knockout mice using the WB method). To avoid potential confusion for readers, we removed Extended Data Fig. 5e (in the original manuscript).

8. Do WT and IFNAR$^{-/-}$ mice express similar levels of ACE2 after Ad5-hACE2 transduction?

In fact, we have previously reported this method of constructing animal models on multiple occasions and validated it in various genetically modified mouse models (including IFNAR$^{-/-}$ and STAT1$^{-/-}$), demonstrating that the Ad5-receptor transduction method is highly stable and reliable (PMID: 24599590, PMID: 32643603, PMID: 36652473).

We evaluated human ACE2 (hACE2) expression in both the presence and absence of SARS-CoV-2 infection following transduction with Ad5-hACE2 in WT and IFNAR$^{-/-}$ mice. No significant differences were observed between wild-type (WT) and IFNAR$^{-/-}$ mice at mRNA expression levels.

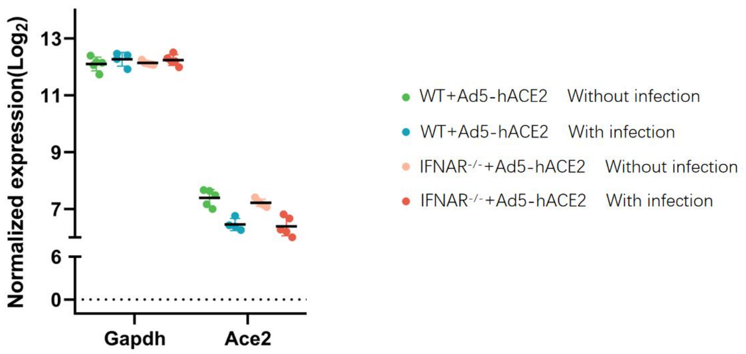


Figure legend: IFNAR$^{-/-}$ and WT were intranasally transduced with 2.5×10^8 FFU of Ad5-hACE2 or Ad5-EV. Five days post transduction, mice were infected with 1×10^5 PFU of SARS-CoV-2 (WT). The lungs were harvested after 2 day post infection and analyzed by mRNA-seq. hACE2 mRNA profiling of homogenized lungs is shown.

To further investigate the ACE2 expression levels following Ad5-hACE2 transduction, we repeated the transduction experiments in both WT and IFNAR$^{-/-}$ mice. The results confirmed that ACE2 protein expression levels were similar between WT and IFNAR$^{-/-}$ mice (as shown below).

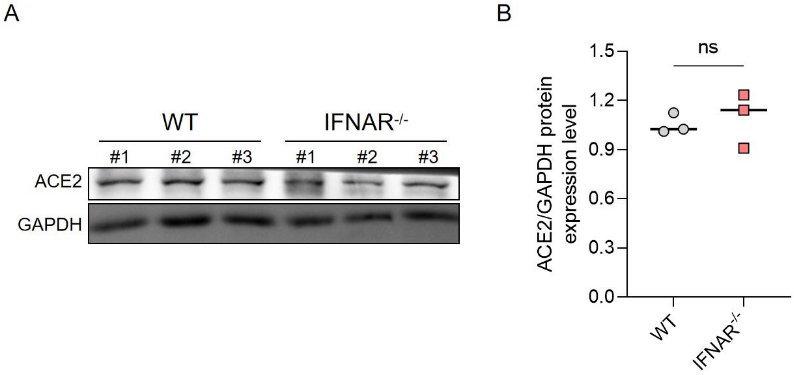


Figure legend: (A) Western blot analysis of human ACE2 expression in WT and Ifnar$^{-/-}$ mouse lungs 5 days after Ad5-hACE2 transduction. (B) The protein levels of human ACE2 in (A) were quantified by grayscale and subsequently subjected to statistical analysis.

9. Some figure legends should provide more experimental details, such as the number of replicates, independent repeats, and the specific meaning of error bars. The English is faulty at some place.

We appreciate the reviewer's valuable feedback. In response, we modified our figure legends and provided more detailed experimental description. Additionally, we revised the overall manuscript to ensure clarity and coherence.

Reviewer #2 (Remarks to the Author):

This manuscript descibes the antiviral activity of O-GalNAc transferase 2 (GALNT2) as interferon regulated gene with antiviral activity against SARS-CoV-2 and influenza A virus. GALNT2 is regulated by type-I IFN both in vitro and in vivo and its depletion and overexpression in vitro and in an AAV overexpression model in vivo, leads to increased or decreased virus growth and pathology, respectively. Following these findings, the authors used a series of in vitro experiments, viral like particles and recombinant SARS-CoV-2 virus to identify S810/813 as one GALNT2 glycosylation site. They found that infection of viruses and VLPs generated in GALNT2 expressing cells is impaired, which is corroborated by cell-cell fusion assays showing reduced fusion of cells in the presence of GALTN2 and spike ofSARS-CoV-2. Most interesting is the identification of a glycosylation site at positions S810/813, mutations rendered viruses insensitive to GALNT2, supporting that the identified sites are relevant for virus growth. Overall the authors propose a model whereby viruses released from cells expressing GALNT2 would be inactive for entering neighboring cells. The authors propose a mechanism of virus inactivation – in response to IFN or more specifically GALNT2 – which would limit virus growth and disease severity.

This is an interesting manuscript describing GALNT2 as antiviral protein and providing a potential mechanism for its activity. The data is supported by in vitro, in cell and in vivo evidence. Most intriguings the notion that GALNT2 expressing cells would release inactive viruses, which are incapable to infect cells, but which may even serve as activators of immune cells in infected tissue. The study thus adds a conceptual advancement on the activity of the innate immune system. My overall enthusiasmis slightly affected by insufficient descriptions of experimental procedures, missing raw or processed data, data not being deployed in public repositories or as supplementary data, which affected reviewing these datasets. However, I support publication of this manuscript since it is an interesting concept and may be an important aspect of antiviral immunity. Additional points need to be addressed, though.

We sincerely thank the reviewer for your insightful comments on our research and for providing valuable and constructive feedback, which has significantly enhanced the quality of our manuscript. In response, we provided comprehensive descriptions of the experimental procedures and all raw data (source data file). The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD051101. The accession number for the Next-gen RNA sequence data reported in this paper is Gene Expression Omnibus (GEO) database: GSE301783. We believe that the quality of our manuscript has significantly improved.

Major points

1. Methods, analysed data and all source data on the presented RNA-seq, LC-MS/MS and glycoproteomics analyses and associated statistical analyses are missing in material and methods or supporting material. Primary data in online repositories is missing.

We appreciate the reviewer's feedback. In the revised manuscript, we provided comprehensive descriptions of the experimental procedures and ensured that all raw data are deposited in public repositories or included as supplementary data (source data file) in our revised manuscript.

2. Statistics - the authors should use SD instead of SEM, please indicate the number and nature of data points that were considered.
We appreciate the reviewer's feedback.
We consulted with a professional statistician and discussed our statistical methods in depth:
- “Report SD” when describing data distribution.
- “Use SEM” when discussing the mean’s reliability or comparing means across studies.
We conducted a thorough review of a substantial number of high-impact journals and observed that in articles reporting results analogous to ours, both SD and SEM methods were employed for data analysis.
As suggested, we used the SD instead of SEM, and we have indicated all experimental details and numbers in the figure legends and methods in the revised manuscript.

3. Figure 1: The authors use an ACE2 expressing AAV vector to transduce IFNAR KO mice with ACE2 and thereby show that a type-I dependent mechanismis restricting SARS-CoV-2 in vivo. I understand that expression of ACE2 in IFNAR KO mice is only a tool to increase infectability and to lead to the identification of GALNT2. However, an alternative interpretation of this experiment could be that the ACE2 overexpressing AAV vector is better infecting IFNAR KO mice (e.g. more ACE2 expression or different distribution) and the described differential infectability in IFNAR KO vs wt mice is driven by differential expression patterns of ACE2. I feel that the authors need to clarify this point. Best would be immunohistochemistry analysis of ACE2 expression in both mouse strains. An additional option would be to plot the ACE2 expression levels in the RNAseq dataset of Fig 1a (which is unfortunately not provided within the current version of the manuscript).
In fact, we have previously reported this method of constructing animal models on multiple occasions and validated it in various genetically modified mouse models (including IFNAR$^{-/-}$ and STAT1$^{-/-}$), demonstrating that the Ad5-receptor transduction method is highly stable and reliable (PMID: 24599590, PMID: 32643603, PMID: 36652473).

Based on our RNAseq dataset (as shown below), we evaluated the expression level of human ACE2 (hACE2) in the presence and absence of SARS-CoV-2 infection following transduction with either Ad5-hACE2 or Ad5-EV. No significant differences were observed in mRNA expression levels between wild-type (WT) and IFNAR$^{-/-}$ mice.

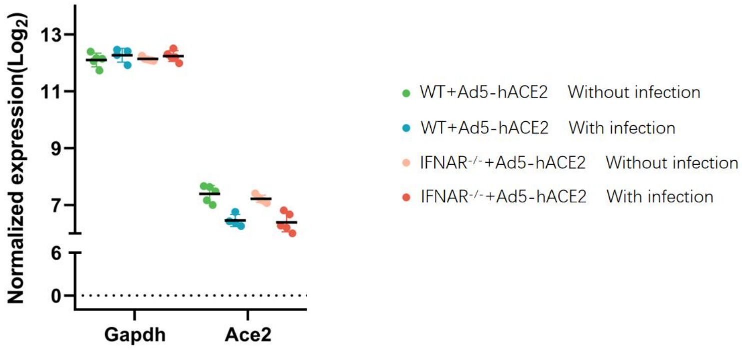


Figure legend: IFNAR$^{-/-}$ and WT were intranasally transduced with 2.5×10^8 FFU of Ad5-hACE2 or Ad5-EV. Five days post transduction, mice were infected with 1×10^5 PFU of SARS-CoV-2 (WT). The lungs were harvested after 2 day post infection and performed mRNA-seq. The hACE2 mRNA profiling of homogenized lungs is shown.

To further investigate the hACE2 expression levels following Ad5-hACE2 transduction, we repeated the transduction experiments in both WT and IFNAR$^{-/-}$ mice. The results confirmed that hACE2 protein expression levels were similar between WT and IFNAR-/- mice (as shown below).

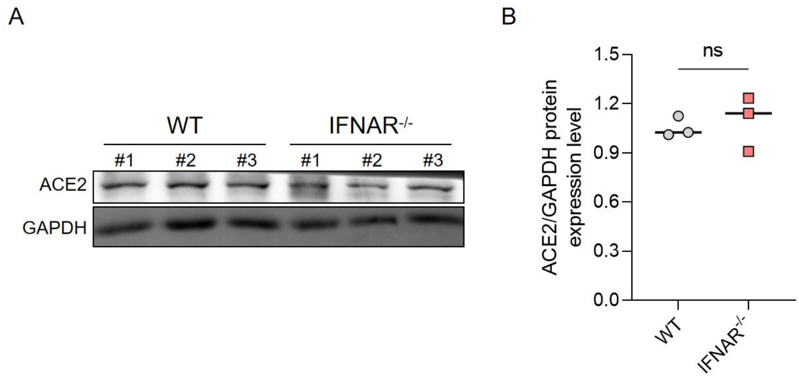


Figure legend: (A) Western blot analysis of human ACE2 expression in WT and Ifnar$^{-/-}$ mouse lungs 5 days after Ad5-hACE2 transduction. (B) The protein levels of human ACE2 in (A) were quantified by grayscale and subsequently subjected to statistical analysis.

4. Figure 1c: The authors are transiently transfecting pcDNA3 based plasmids into Huh7 cells and infect these cells with SARS-CoV-2 for N staining.I am not a big fan of such assays since heterogeneity of the transfection efficacy of the individual plasmids and the differential strength of overexpression of individual proteins can lead to heterogenous protein expression, differentially affect cell viability and thereby general viral propagation. Thus this experiment may be limited in its interpretability. Moreover, overexpression of individual proteins (such as IFIH1) may change the overall antiviral state of the cells without much specificity to the given infection. This concern may be reflected by the fact that that all 28 tested ISGs are somewhat antiviral… it is simply uncommon that all transcriptionally regulated genes have antiviral activity. I suggest to use stably transduced cells for Figure 1c with control for gene expression or to remove these data.

We are very grateful to the reviewer for your valuable suggestions.

Firstly, we previously compared GALNT2 protein expression levels in stably transfected cell lines (OE-GALNT2) (Figure A, left) and transiently transfected (plasmid-GALNT2) (Figure A, right；the intensity of GALNT2 was qualified and refer to Question 5 of Reviewer 2). As shown below, the GALNT2 protein expression level achieved through transient transfection closely resembles the expression levels observed under the interferon stimulation and viral infection conditions (Figures C-D). In contrast, the expression level of GALNT2 in the stably transduced cell line is significantly weaker. Furthermore, upon analyzing the infection dynamics in both cell types, it was evident that for both the SARS-CoV-2 and influenza virus (PR8), the inhibitory efficiency of transiently transfected cells significantly surpassed that of stably transfected cell lines, demonstrating the inhibition is dependent on the dose of GALNT2 (Figure E-F).

In GALNT2-stable cell lines, there is a gradual decrease in the protein expression levels with the increasing number of passages, and lower antiviral activity compared with transiently transfected cells, possibly because the lentivirus transduction might have induced a slight level of innate response already.

When a virus infects a cell, it can rapidly induce the production of interferons and a large number of ISGs. We found that the protein expression level of ISGs produced by transient transfection is relatively close to that of ISGs induced by viral infection. Therefore, we would like to clarify that using transient transfection to study ISGs is a relatively classical method, with numerous high-impact publications supporting this methodology. (PMID: 33113352, PMID: 34413236). In addition, we did not solely rely on overexpression data. We further confirmed the phenotype and working mechanism of GALNT2 under GALNT2 knockout and knockout/rescue conditions in cell lines. The restriction function of GALNT2 was also validated in a GALNT2 knockout mouse model. Overall, although GALNT2 was initially identified through overexpression screening, we thoroughly employed multiple physiological approaches to confirm its phenotypes and mechanisms.

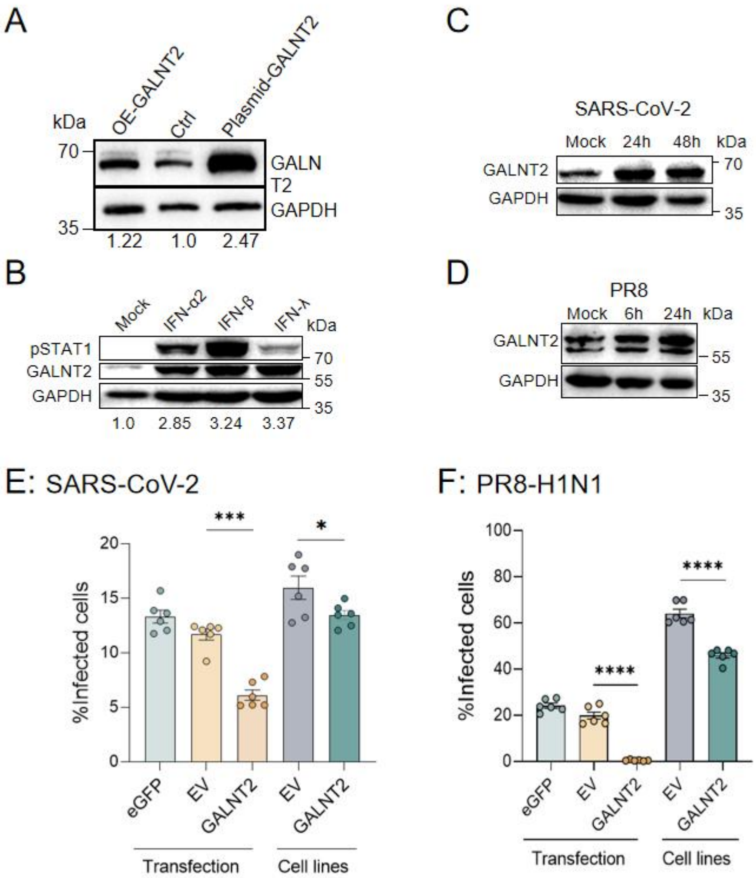


Figure legend: (A) GALNT2 protein expression levels in GALNT2-Stably cells (left), WT (middle), and transiently transfected with GALNT2 plasmids (right). (B)THP-1 cells were treated with 5 μg/mL IFN-α2, IFN-β, or IFN-λ1. After 24h post treatment, GALNT2 expressions in cell lysates were analyzed by western blot. (C)Huh7 cells infected with WT SARS-CoV-2 (MOI=0.1) for 0 (mock), 24 and 48 h. GALNT2 expression levels were determined by western blot analysis. (D)A549 cells were infected with PR8 H1N1 at MOI of 0.01. Cell lysates were harvested at 6 and 24 h.p.i., respectively. GALNT2 and pSTAT1 protein expression levels were detected by western blot. (E) Huh7 cells were transfected plasmids encoding with eGFP, empty vector (EV), or GALNT2. After 24h post transfection, along with Huh7 cells stably overexpressing EV or GALNT2, cells were infected with SARS-CoV-2 (MOI=0.01). Viral infection efficiency was assessed by IFA 24 hours post-infection. (F) A549 cells were transfected plasmids encoding with eGFP, empty vector (EV), or GALNT2. After 24h post transfection, along with Huh7 cells stably overexpressing EV or GALNT2, cells were infected with PR8 (MOI=0.01). Viral infection efficiency was assessed by IFA 24 hours post-infection.

On the other hand, the reviewer highlighted that the majority of the 28 ISGs we screened exhibit antiviral activities compared with EV and unrelated protein (eGFP) controls. In addition, we have previously employed empty vector and unrelated proteins (e.g., eGFP) as negative controls in our experiments already, none of which exhibited any antiviral activity. These results might be attributed to our targeted screening of ISGs from a large pool of genes. Among these 28 identified genes, 24 of these ISGs have been reported with antiviral functions, while the antiviral activity of 4 ISGs (GALNT2, G6PC, PLEKHA4, SECTM1) have not yet been reported. Therefore, we focused on these four genes and confirmed their antiviral activity in cells

(Extended Data Fig. 1c). We will further examine the function of the remaining three genes in cells and mouse models in future studies.

In response to the concerns of the reviewer, we have previously employed empty vector and unrelated proteins (e.g., eGFP) as negative controls in our experiments already, and we now included these data in Figure 2 and throughout all transient expression experiments. Additionally, we included the transfection efficiency (Question 5 of Reviewer 2) and cell viability (Question 8 of Reviewer 2) of these experiments in our revised manuscript (Extended Data Fig.1h-k). As suggested by the reviewer, we have relocated the relevant experiments, including the process of screening of GALNT2, to the supplementary figures (Extended Data Fig.1).

5. This paper generally employs diverse means of protein overexpression using plasmid transfection, stable cell lines, lentiviral transduction. It is often unclear from figure legends which type of overexpression has been used. This is an important variable in the given experiments, please clarify this for every experiment shown. For some transient plasmid-transfection mediated overexpression experiments it would be important to show transfection efficacy (% of infected cells) since untransfected cells would influence some of the assays.

We sincerely thank the reviewer for your insightful suggestions. As suggested, we have included the details for all overexpression experiments in the figure legends.

We also confirmed the plasmids transfection efficiency in multiple conditions. The transfection efficiency of the GALNT2-HA transient transfection using Lipofectamine 3000 in HEK293T and Huh7 cells were analyzed by the WB (Figure A), IFA (Figure B) and FACS (Figure C) analysis using an anti-HA tag or anti-GALNT2 antibody.

In a separate experiment, we also transiently overexpressed eGFP in the Huh7 cell line that we were using. 24 hours after transient transfection, expression of eGFP in Huh7 cells was captured using a fluorescent microscope and further analyzed by FACS. It shows that the transfection efficiency of eGFP in Huh7 cells was at least up to 77.7% (Figure D). We included these data in the revised manuscript (Extended Data Fig.1h-i)

Each of our experiments used empty vector (EV) and unrelated proteins (eGFP) as controls. We have clarified and expanded our methods and figure legends, particularly for the overexpression experiments. As suggested by the reviewer, we have relocated the relevant experiments, including the transfection efficiency of GALNT2 to Extended Data Fig.1h-i.

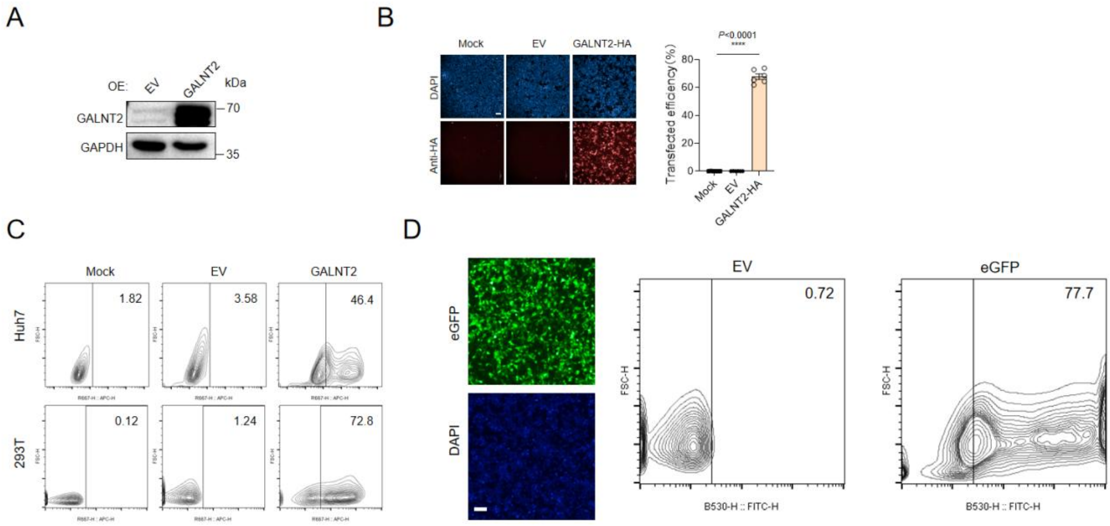


Figure Legend: (A) Huh7 cells are overexpressed of GALNT2 (EV as the control), and GALNT2 protein expression level was detected at 24 hours post transfection by western blots. (B) Huh7 cells were transiently transfected with Lipofectamine 3000 packaged with pcDNA3.1 empty vector (EV), or plasmids encoding the GALNT2-HA for 36 hours. Cells were then fixed with 4% paraformaldehyde and stained using Anti-HA-tag antibodies conjugated with Allophycocyanin (APC). Cells were analyzed by IFA. (C) Huh7 and HEK293T cells were seeded in flat bottom 24-well plates at 2 x 10^5 and 5 x 10^5 cells /mL respectively overnight. 300 ng pcDNA3.1 plasmids encoding EV, GFP or GALNT2-HA were packaged in Lipofectamine 3000 (Life technologies) and transiently transfected for 24 hours. HEK293T and Huh7 cells in the 24-well plates were then detached using trypsin-EDTA and centrifuged at 600 g for 4 min. Cells were then resuspended and chemically fixed in 4% PFA for 15 min at room temperature. For staining of the GALNT2-HA tag, cells were further permeablized in 0.1% Triton buffer for 10 mins, before blocked in 1% BSA PBS for 30 min. APC-conjugated mouse Anti-HA tag antibody (901523, Biolegend) was then added to cell suspension for the staining of GALNT2 for 1 hour, before washed three times with PBS. Flow cytometry was carried out on a Novocyte Advanteon (Agilent), subsequent gating and data analysis was performed using the FlowJo 10.4 (LCC).

6. The authors describe GALNT2 as yet unstudied ISG. Therefore additional information and relation to canonically used ISGs would be very interesting to readers. The manuscript may benefit from additional information on tissue distribution and its already known regulation in diseases – such data can easily be provided by consulting public databases.

We thank the reviewer for this insightful suggestion. We have added additional information regarding GALNT2's tissue distribution, its comparison with canonical ISGs, and its regulation in COVID-19.

For tissue distribution analysis, we downloaded RNA and protein expression data across human tissues from the Human Protein Atlas (https://www.proteinatlas.org). At the RNA level, GALNT2 is highly expressed in the pancreas, liver, and skeletal muscle (Figure a), whereas at the protein level, high expression is observed in the pancreas, lung, kidney, skin, testis, and placenta (Figure b). This expression pattern is distinct from those of canonical ISGs such as OAS1 and Mx1, both at the RNA and protein levels.

To investigate the regulation of GALNT2 in COVID-19, we analyzed single-cell RNA-seq data from bronchoalveolar lavage fluid (BALF) samples (the same dataset as used in the main text), and proteomics data from a published study (PMID: 33503446). In COVID-19 patients, GALNT2 is significantly upregulated at both the RNA (Figure c-d) and protein levels (Figure e) in multiple cell types and tissues. Notably, the upregulation pattern of GALNT2 differs from that of OAS1 and MX1, suggesting a distinct regulatory response during SARS-CoV-2 infection.

We included these data in Extended Data Fig. 3 in the revised manuscript.

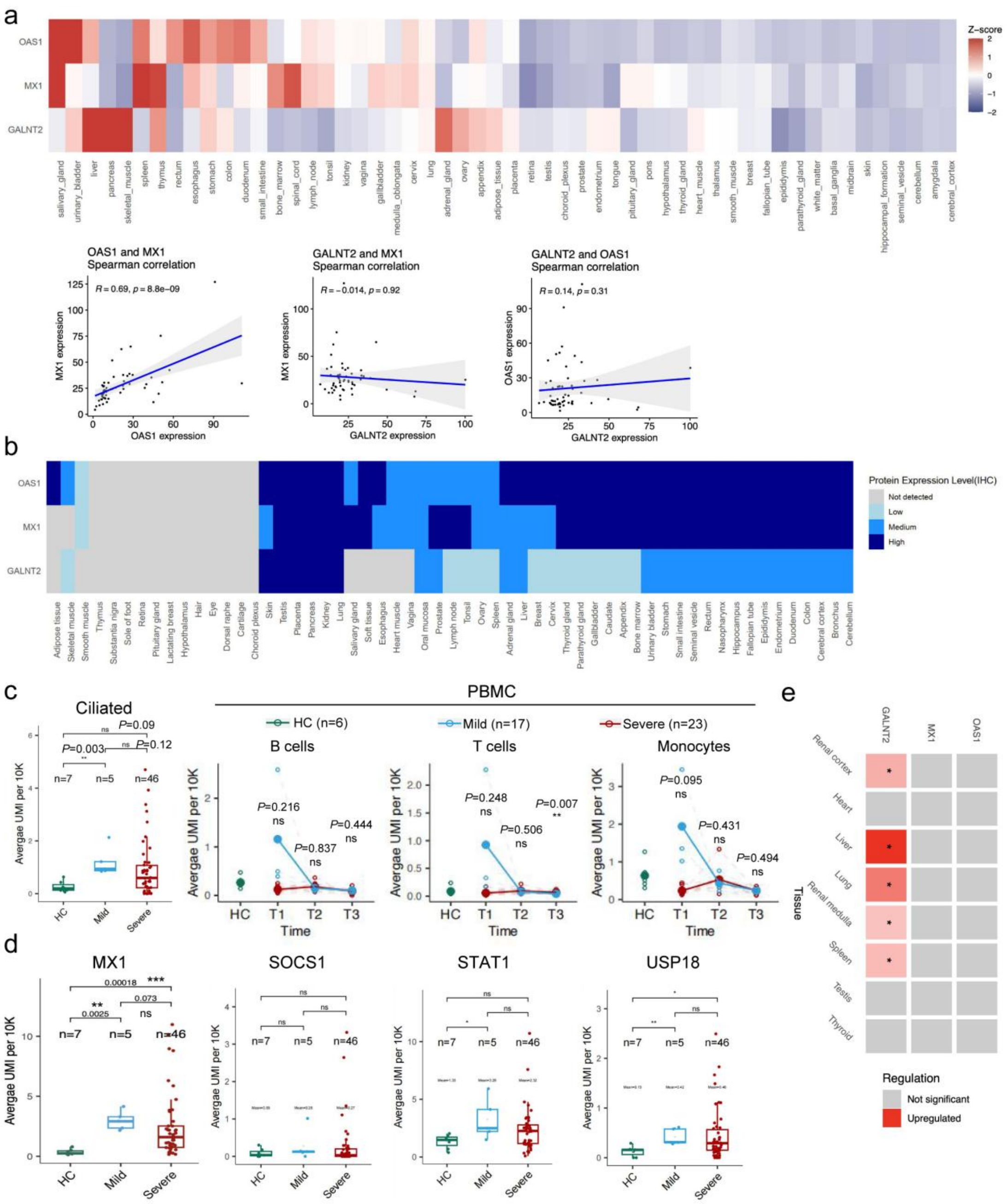


Figure legend: (a) RNA expression patterns of OAS1, MX1, and GALNT2 across human tissues (data from the Human Protein Atlas; https://www.proteinatlas.org), visualized as z-score normalized heatmaps (red: higher expression, blue: lower expression). Scatter plots below show pairwise expression correlations between genes, with locally estimated scatterplot smoothing (LOESS) fits (blue line) and 95% confidence intervals (gray shading). Pearson correlation coefficients and p-values are indicated. (b) Protein-level expression of OAS1, MX1, and GALNT2 across human tissues, from the Human Protein Atlas. Expression levels are categorized as not detected(grey), low (light blue), medium (blue), or high (dark blue) by HPA database. (c) Left: Comparison of OAS1 expression in ciliated epithelial cells from bronchoalveolar lavage fluid (BALF) across COVID-19 severity groups based on scRNA-seq data. Each dot represents an individual; the y-axis shows average OAS1 expression per individual. Right: Longitudinal OAS1 expression in peripheral blood mononuclear cell (PBMC) subsets (B cells, T cells, and monocytes) across healthy controls (HC) and patients with mild or severe COVID-19. Open dots represent individual-level averages; solid dots represent group means. Time points: T1 (acute phase), T2 (conversion), T3 (recovery; see Methods for definitions). Samples from the same individual are connected by dashed lines. P values were calculated using Wilcoxon signed-rank test (left) or Student's t-test (right). (d) Comparison of MX1, SOCS1, STAT1, and USP18 expression in ciliated BALF cells across clinical severity groups. Each dot represents one individual; the y-axis shows average gene expression. P values by Wilcoxon signed-rank test. (e) Summary of differential protein expression for the indicated genes across tissues in COVID-19 patients versus controls, based on published proteomics data (PMID: 33503446). Red indicates significant upregulation (FDR < 0.05); gray, not significant.

7. Figure 1e/f shows the dynamics of GALNT2 – how does this compare to canonical ISGs such as OASs, IFITs or STATs or USP18, SOCS (which are negative regulators expressed after the first wave of ISGs)? To me the expression of GALNT2 appears to to rise quite late in wt cells. Also, please indicate which method was used for quantification and show the raw data (western blots?) – this is not described. Similarly, it would be nice top lot the dynamics of expression data of known ISGs in the scRNAseq experiment of Fig 1i/j (e.g. as extended data). I acknowledge that the authors profile wt and GALNT2 KO cells using transcriptomics, proteomics and glycoproteomics. I feel that characterization of the same cells in infected conditions could be informative for readers to understand the function of GALNT2 in infected conditions.

As recommended by the reviewer, here we compared the protein expression dynamics of GALNT2 with STAT1 (as shown below) in mice. We included these data in the Extended Data Fig. 1e.

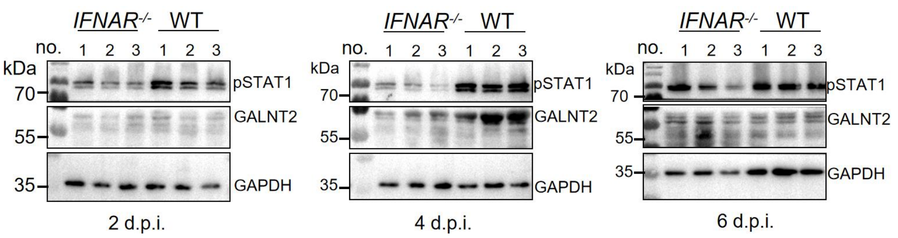


Figure legend: Western blot analysis of mouse GALNT2 and STAT1 protein expression in SARS-CoV-2-infected WT and *IFNAR*$^{-/-}$ mouse lungs at 2, 4, 6 d.p.i. after Ad5-hACE2 transduction.

Based on our scRNA-seq data (as shown below), we have already analyzed OAS1 expression in different cell types. Intriguingly, OAS1 expression levels were markedly increased in ciliated cells and only a slightly upregulated trend in PBMC cells at initial infection period (T1) in mild COVID-19 patients, indicating different antiviral mechanisms between OAS1 and GALNT2. We included these data in the Extended Data Fig. 3c.

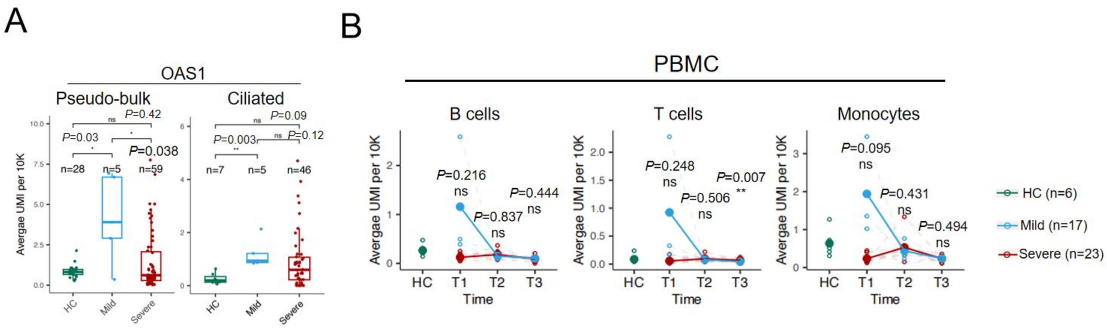


Figure legend: Comparison of OAS1 expression in BALF (A) and PBMC (B) samples using scRNA-seq. Each dot represents an individual, and the y-axis represents the average expression level of OAS1 in a specific cell type from that individual.

Further, we analyzed and showed the expression pattern of IFIT1, STAT1, USP18, SOCS1 and MX1 in the BALF cells and PBMC. In fact, our findings indicate that the expression patterns and kinetics of these canonical ISGs can differ significantly. This variation is intricately linked to the intrinsic properties and functional roles of each gene, as well as the diverse antiviral mechanisms they engage in. Moving forward, we will continue to closely monitor these intriguing observations. We included these data in the Extended Data Fig. 3d.

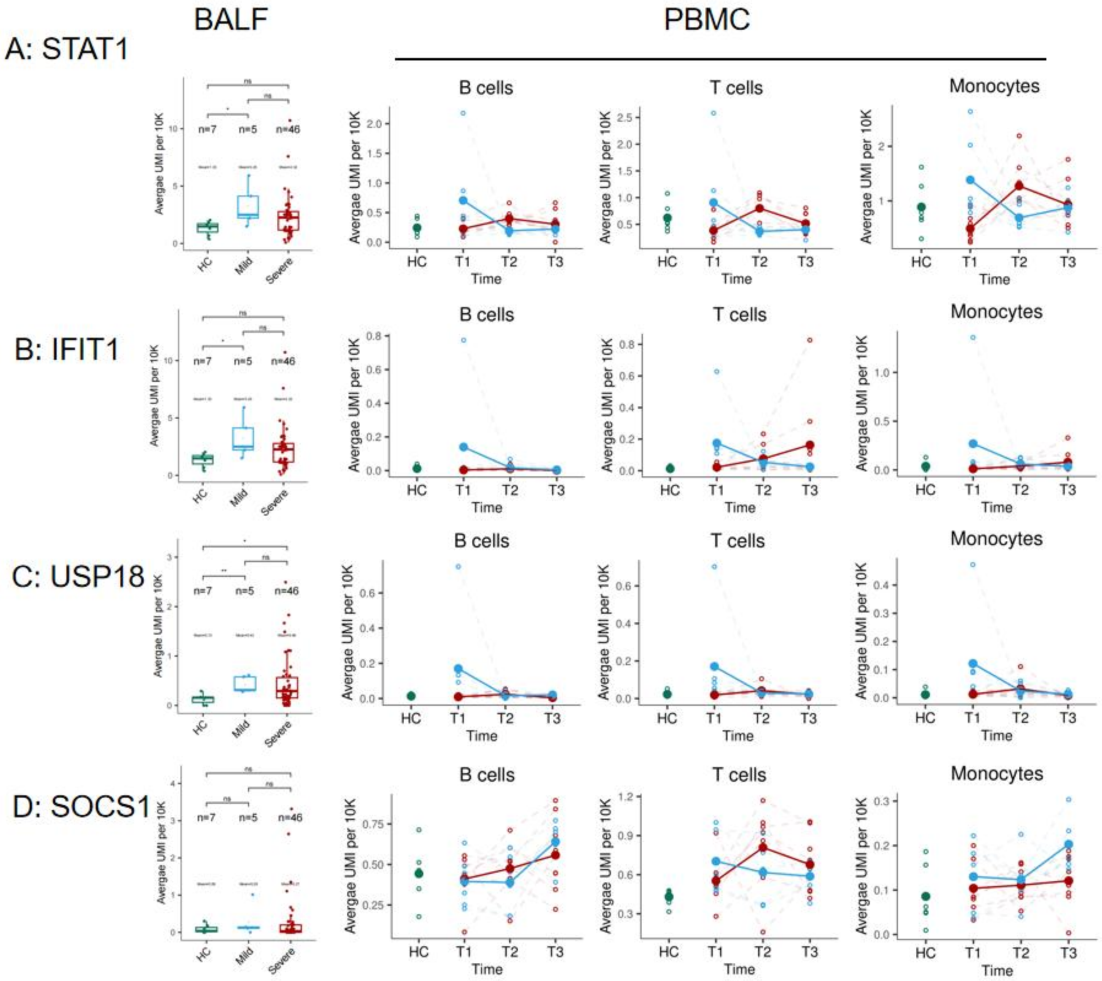


Figure legend: Comparison of STAT1 (A), IFIT1 (B), USP18 (C) and SOCS1 (D) expression in BALF and PBMC samples using scRNA-seq. Each dot represents an individual, and the y-axis represents the average expression level of OAS1 in a specific cell type from that individual.

In our study, we profiled wild-type (WT) and GALNT2-overexpressing cells using transcriptomics, proteomics, and glycoproteomics. The rationale for conducting multiple omics analyses was to rule out the possibility of indirect effects:

1) Other functional antiviral ISGs might be influenced by GALNT2.

2) Pathways regulating innate immunity and other known pathways directly involved in viral infection might be modulated by GALNT2.

Based on the omics results, we did not observe such side effects, which further confirms that *O*-linked glycosylation of the spike protein is the primary target of GALNT2.

Lastly, we tried to profile wt and GALNT2-stably cells using transcriptomics after infection (as shown below). However, under infection conditions, both cell types were infected with the virus and exhibited widespread gene induction. Under these conditions, it is challenging to determine whether the differences in gene expression are attributable to GALNT2 overexpression or varying levels of viral replication (as GALNT2 overexpression has been shown to attenuate viral replication). From an alternative perspective, the overexpression of GALNT2 might have no discernible direct effect on host proteins, thereby substantiating our conclusion.

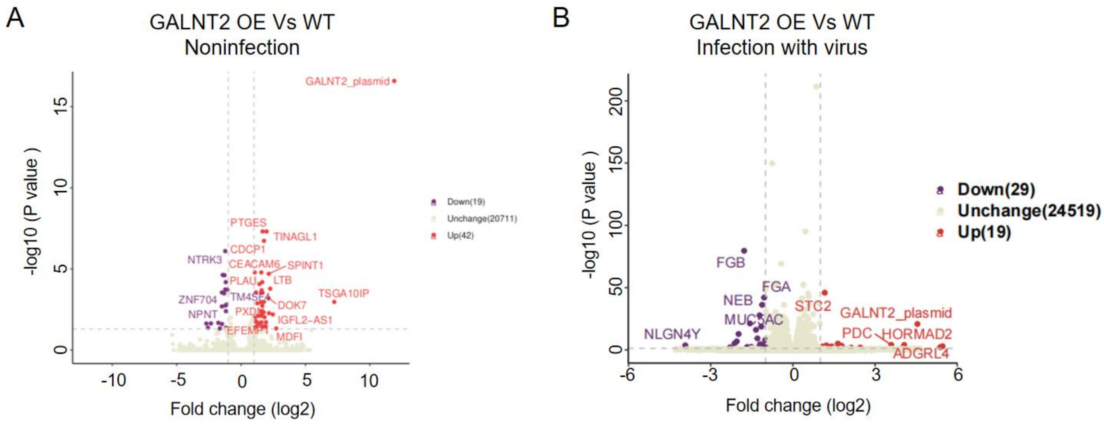


Figure legend: A549 cells overexpressing GALNT2 or not were harvested for mRNA-seq analysis without or with influenza A virus (PR8-H1N1) infection at 24 hours post infection. Greater than 2-fold change in abundance upon GALNT2 overexpression and lower than 0.05 adjusted p value were set as the threshold criteria for GalNAc-T2-regulated proteins.

8. Figure 1d, Figure 2: Please indicate how the overexpression cells were generated – are these stable cells? Does GALNT2 expression affect cells overall? The DAPI staining in Figure 1d looks more dull – does this reflect reduced cell growth? It is important to characterize the used cells by testing the cellular metabolism (e.g. ATP levels by Cell titre glow or similar) and evaluate the growth kinetics of these cells over time. It is also not clear from the figure legends how statistics were determined. These are representative experiments of repeats and the small error bars indicate that these are based on some technical (how many?) replicates. Statistics and average values considering all biological repeats would be reassuring to evaluate the effect.

We added the experimental and statistical analysis details for GALNT2 overexpression cells, as suggested by the reviewer. No significant changes were observed in the GALNT2-expressing cells compared to the control cells. Additionally, we provided quantification data monitoring cellular ATP levels to further confirm the effect of GALNT2 on cell viability in A549 (Figure A) and Huh7 (Figure C) cells. In fact, our findings indicate that regardless of whether GALNT2 was overexpressed (transiently transfection), it exerted no notably significant effect on cell viability, as well as cell growth (Figure B and D) .

Additionally, in the infection experiments, cells were consistently plated at a predetermined density (approaching 80% confluence) and subsequently subjected to viral infection after a 16-hour incubation period. Consequently, under these experimental conditions, the impact of cell activity and growth could be effectively disregarded.

We further explored the impact of GALNT2 overexpression and concluded that there is no major impact on the host cell by measures of transcriptomics/proteomics and glycoproteomics. All data demonstrated that GALNT2 overexpression might not influence cellular metabolism and cell growth.

As suggested, we included these data in the Extended Data Fig. 1j-k in the revised manuscript.

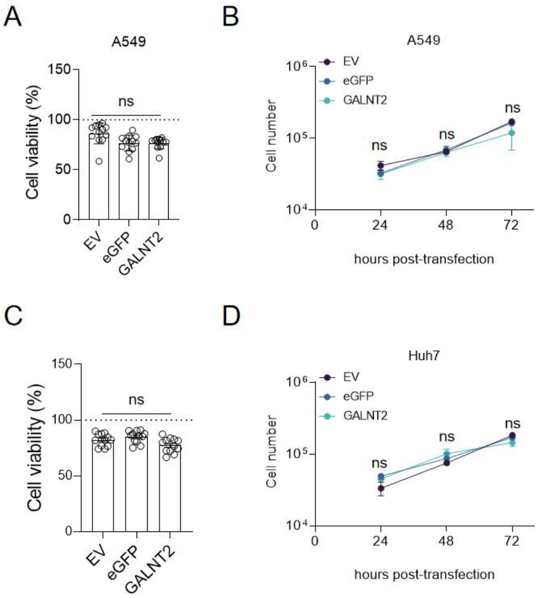


Figure legend: (A)Wild-type A549 (A549 WT), empty vector-overexpressing (A549 OE-EV), eGFP-overexpressing (A549-OE-eGFP), GALNT2-overexpressing (A549 OE-GALNT2), cells were seeded in 96-well plates at 3.5×10^4 cells/well. WT A549 cells were seeded at 96-well plate (3.5×10^4 cells/well) and transfected with empty vector (EV), eGFP, or GALNT2 plasmids for 24h. Cell viability was measured 24 hours post-seeding (cell lines) or post-transfection (transfected cells). Data represent mean ± SD of three independent experiments. (B) A549 OE-EV, A549 OE-GALNT2, A549 OE-eGFP cells were seeded in 24-well plates at 1×10^4 cells/well. Cells were trypsinized and counted at 24, 48, and 72 hours post-seeding. Three independent experiments performed. (C-D) Similar experiments were conducted in Huh7 cells.

9. Figure 2c (and extended Figure 3): An essential control of transfecting an unrelated protein is missing to exclude cellular resistance induced by the transfection procedure. It would be further essential to know transient transfection efficacy (of GALNT2 and currently missing control) in these cells.
As suggested, we have previously employed empty vector and unrelated proteins (eGFP) as negative controls in our experiments already and we now included these data in the revised manuscript (Fig. 2 and Extended Data Fig. 5d-e). Further, we incorporated transient transfection efficacy of GALNT2 in Extended Data Fig. 1h-i. Lastly we also provided quantification data monitoring cell viability and cell growth in GALNT2 overexpression cells (Extended Data Fig. 1j-k) as shown above. (Refer to Questions 5 and 8 of reviewer 2).

10. I am a bit surprised by the expression of GALTN2 in THP-I cells by IFN-lamda treatment. THP-I cells appear not to express IFN-lambda receptor (also evidenced by

the reduced STAT1 levels) – what is inducing GALTN2 in these cells? Can the authors please plot for an alternative ISG in this cell type?

We reconfirmed that the previous report supports our observation. It demonstrates that THP-1 cells exhibit low expression levels of IFNLR1 (IL28RA) while maintaining normal expression levels of IFNLR2 (IL10RB) (PMID: 32353085). This observation may explain why IFN-λ stimulates GALNT2 expression.

Further, we also analyzed mRNA expression levels of other ISGs in THP-1 cells after IFN treatment at 6h post stimulation and we found all these known ISGs could be stimulated (as shown below).

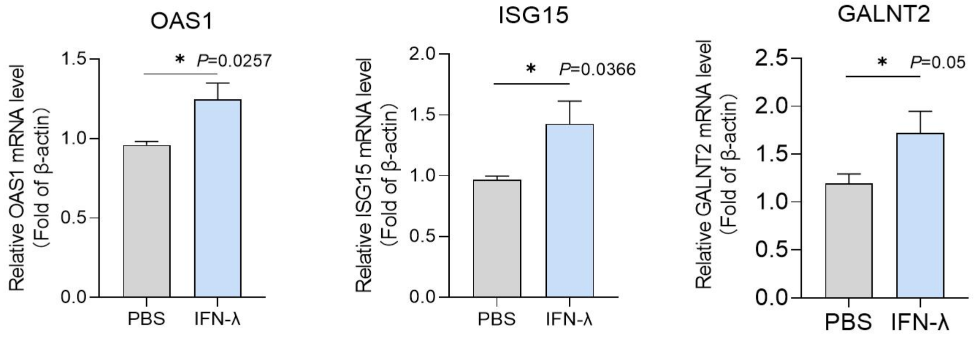


Figure legend: THP-1 cells were seeded in a 24-well plate and treated with either PBS or 500 ng/mL IFN-λ in the culture supernatant. The mRNA expression levels of GALNT2, OAS1, and ISG15 were quantified by qPCR 6h post-stimulation.

11. Figure 1f: figure legend mentions the expression of STAT1 in IFNAR KO mice, but these blots are not shown. It would be an interesting control, though.

As suggested, we provided STAT1 data in the revised manuscript (Fig. 2f).

12. Extended Figure 5e, g – it is unclear to me what "fold change" means in this case… which conditions were compared? It would maybe be better to plot delta-CT values vs a housekeeping gene.

WT and GALNT2$^{-/-}$ C57BL/6 mice were intranasally infected with 350 FFU of PR8 H1N1. Lungs were harvested at 3 d.p.i. Then we detected the GALNT2 and IL-1β expression levels compared with GAPDH in WT and GALNT2$^{-/-}$ mice.

Fold change indicates the comparison of GALNT2 with GAPDH by qPCR assay with high sensitivity or some extent of background readings (Extended Data Fig 5e in the original manuscript). In fact, we confirmed the GALNT2 knockout in mice by intranasally stimulation with 2 μg of IFN-α2 and were unable to detect GALNT2 expression via Western blot (Extended Data Fig. 5c in the original manuscript). To avoid potential confusion for readers, we removed original Extended Data Fig. 5e.

As suggested, we modified the figure legend to make it clear.

13.Extended Figure 5f –the authors mention it in the text that GALNT2 depletion affected lung pathology and show one example of wt and GALNT2 KOs – it would be nice if the authors could this across the different mice tested. I understand that the shown lungs are very supportive there.

We appreciate the reviewer's valuable feedback.

We analyzed all five mice involved in the infection experiment and observed that all GALNT2 KO mice displayed more prominent pathological features, such as inflammatory infiltration, thickening of the alveolar walls, and hemorrhage. In the revised manuscript, we have included all results for conciseness (Extended Data Fig. 7f).

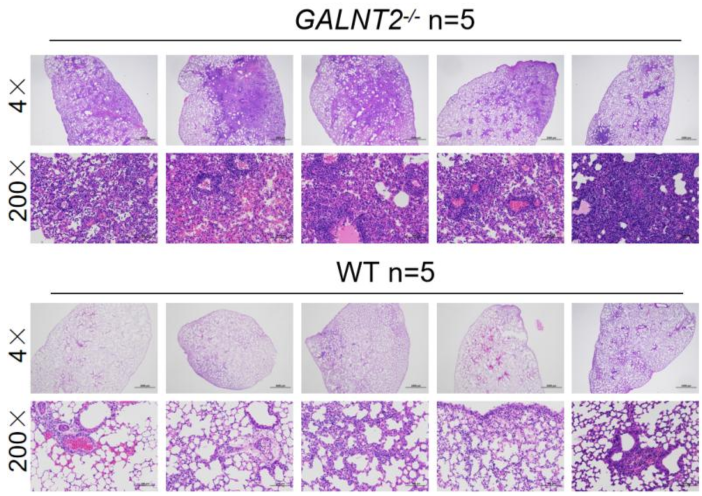


Figure legend: WT (n=5) and GALNT2$^{-/-}$ (n=5) C57BL/6 mice were intranasally infected with 350 FFU of PR8 H1N1. Lungs were harvested at 3 d.p.i.. HE staining of the infected lungs of WT or GALNT2$^{-/-}$ C57BL/6 mice were shown.

14. Extended Figure 6c- I (and other in vivo data): Currently a variable number between 3 and 6 data points are plotted (though often coming from one experiment). Please plot not detected values (you can indicate detection limits accordingly). It would further be nice to indicate the number of mice that were assessed in these experiments in Figure legends.

We thank the reviewer for this insightful suggestion. We have made the suggested modifications in Extended Data Fig. 8c in the revised manuscript and modified the figure legend (lines 1830-1831) to make it clear.

15. Figure 3d, e; Extended Figure 6g-k: Can the authors exclude that GALNT2 expression affects subsequent infection with AAV-ACE2? ACE2 is required for SARS-CoV-2 infection and impaired expression may affect this in vivo model. For the interpretability of the data, potential impairment of AAV-ACE by GALNT2 expression would be important to be evaluated.

We appreciate the reviewer's feedback. To further detect the ACE2 expression levels after Ad5-hACE2 transduction, we repeated the transduction experiments. The results confirmed that GALNT2 overexpression does not influence the ACE2 expression levels. Lastly, we also confirmed the ACE2 expression levels independent on GALNT2 overexpression in Huh7 cells.

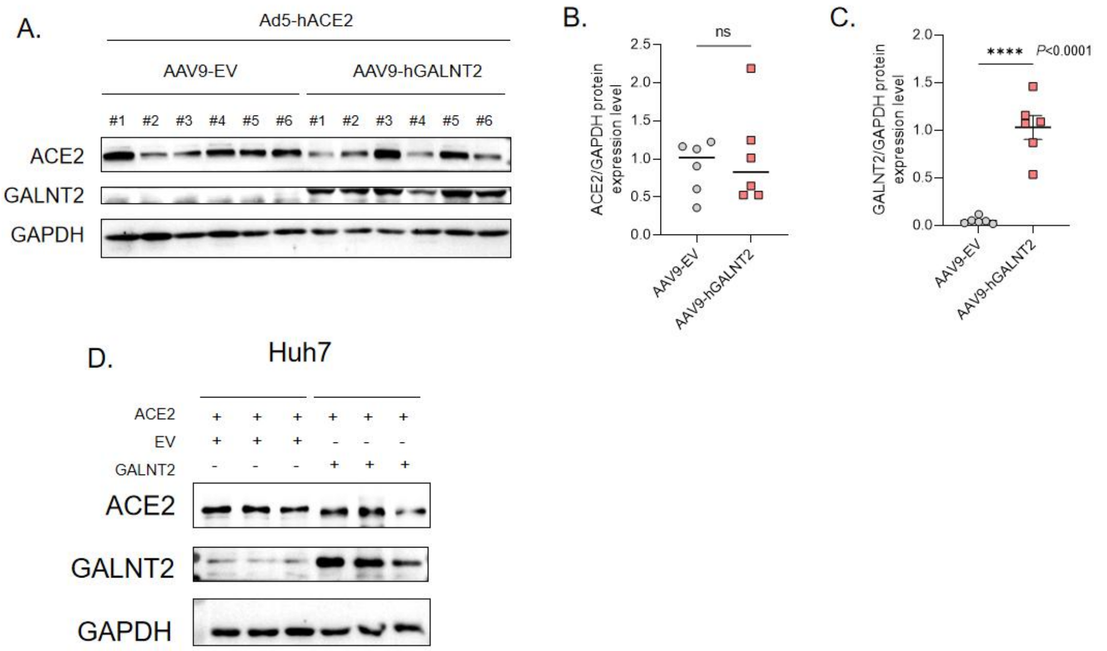


Figure legend: (A) Western blot analysis of human ACE2 (hACE2) and GALNT2 expression in WT mouse lungs following 14 days of AAV9-EV (n=6) /hGALNT2 (n=6) transduction and subsequent Ad5-hACE2 transduction for 5 days. (B)The protein levels of human ACE2 in (A) were quantified by grayscale and subsequently subjected to statistical analysis. (C)The protein levels of GALNT2 in (A) were quantified by grayscale and subsequently subjected to statistical analysis. (D)Huh7 cells were co-transfected with either empty vector (EV) or GALNT2 along with hACE2. Protein expression levels of ACE2 and GALNT2 in cell lysates were assessed by Western blot analysis.

16. Extended Figure 8a: The authors mention that the plaque size of viruses derived from GALTN2 overexpressing cells is smaller. I do not understand why this is the case – would the cells used for plaque assay not promote expression of viruses? I.e. if a virus infects the cells, should it not be propagated similarly, independent of the virus-producing cell in the cell monolayer?

The formation of a viral plaque is an important phenomenon during influenza A virus and SARS-CoV-2 virus infections. Plaque size is commonly employed to assess viral infectivity and fusion capability. Due to the restriction imposed by carboxymethylcellulose (CMC), the release and diffusion of progeny virus for infecting distant cells are inhibited, allowing spread only through spike-mediated viral-cell fusion to neighboring cells, thereby resulting in plaque formation.

SARS-CoV-2 spike is known to be firstly processed at the S1/S2 cleavage site by proprotein convertase furin during viral assembly, while TMPRSS2 makes additional cleavages at spike S2' site during ACE2-dependent viral entry and virus-cell fusion. We demonstrated that GALNT2 *O*-glycosylates S810/813 immediate upstream of the S2' cleavage site and inhibits spike-mediated virus-cell fusion. Since it affects the viral entry process, the plaque size of viruses derived from GALTN2 overexpressing cells was smaller.

17. It is very interesting that influenza HA is inhibited by GALNT2. Does this also apply to HA serotypes with a polybasic cleavage site, such as HAs derived from avian influenza viruses?

We have evaluated the fusogenic activity of the hemagglutinin (HA) protein from the H5N6 (A/Sichuan/06689/2021(H5N6) or H7N9 (A/Guangdong/Th008/2017) strains of avian influenza virus (high pathogenic AIVs with polybasic HA cleavage site) using the cell-cell fusion models. HEK293T cells were co-transfected with the AIV HA, GFP and empty vector (EV) or GALNT2 for 24 hours, before stimulated with pH=5 HEPES buffered saline solutions. After 10 mins stimulation, cells were re-perfused with complete growth media for 2 hours before green fluorescence images were captured and the size of individual syncytium in stimulated cells were quantified by ImageJ. The results demonstrated that cell-cell fusion mediated by the H5 or H7 HA at pH 5 conditions was significantly reduced by the GALNT2 expression. We incorporated these data into Extended Data Fig. 12e-f.

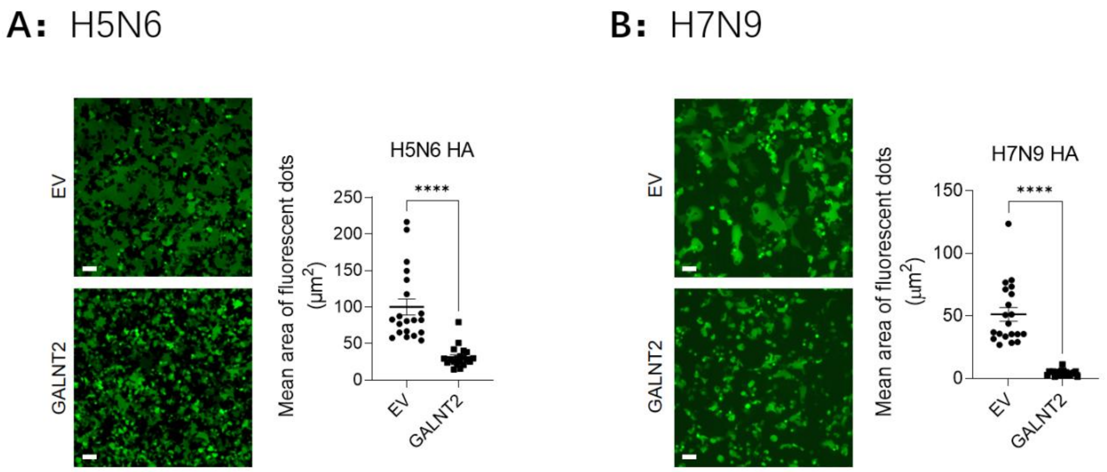


Figure legend: GALNT2 inhibits the fusogenic activity of the hemagglutinin (HA) protein from the avian influenza virus strain H5N6 with polybasic cleavage sites. Fluorescent images of the GFP reporter expression captured from HA (H5N6 and H7N9) and GFP co-expressing HEK293T cells in the presence of EV or GALNT2 for 24h. Cells were then stimulated by a pH5 HEPES-buffered saline for 10 min at 37ºC. Growth medium were replaced and cells were cultured further for 2 h. Images were representative of 3 individual repeats, scale bars are indicative of 50 μm. Individual syncytium derived from HA-mediated cell-cell fusion was quantified as mean area of fluorescent dots in ImageJ.

18. The identification of the glycosylation site at positions S810/813 is very interesting and well done. Particularly growth assays in presence of GALNT2 overexpression are supporting this story. Since GALNT2 has been reported to be quite promiscuous in activity, it would be very reassuring if the glycosylation assay shown in figure 4g was repeated with the S810/813 mutant(s) viruses. What is the occupancy of glycosylation on virus particles that are released from wt and GALTN2 KO and IFNAR or STAT1 KO cells? One would expect that this glycosylation pattern is prominent in wt cells but not in cells lacking GALNT2 or IFN signaling.

We sincerely appreciate the highly professional advice you have provided.

As suggested by the reviewer, we have performed HPA pull-down assays using SARS-CoV-2 pseudoviruses packaged using the WT and the *O*-glycosylation-deficient S810/813A spike mutants. In the presence of GALNT2, we found that interaction of HPA-biotin with the S2 from the S810/813A spike mutant was significantly reduced when compared to the WT. It was demonstrated that the mutation of S810/813A significantly reduced the *O*-glycosylation of the spike protein. We also analyzed the *O*-glycosylation of the spike protein on virions in the presence or absence of GALNT2. The results confirmed that GALNT2 not only mediates *O*-glycosylation of intracellular spike (Fig. 4g-h) but also regulates the occupancy of *O*-glycosylation on virions of the WT spike, whereas this effect was significantly diminished for the S810/813A mutant (as shown in the figure below).

We included these data into the Extended Data Fig. 10h to strengthen our conclusions that GALNT2-mediated *O*-glycans are present on the virion spike S810/813 residues.

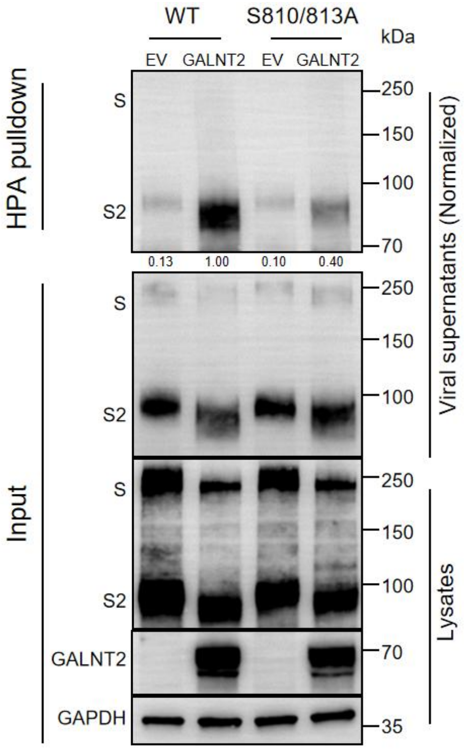


Figure legend: Streptavidin pulldown of HPA-biotin mixed with SARS-CoV-2 pseudoviral particles incorporated with SARS-CoV-2 WT or S810/813A spikes. Supernatants containing the SARS-CoV-2 pseudoviral particles derived from EV or GALNT2-expressing HEK293T cells were collected and verified for SARS-CoV-2 spike incorporation and assembly of VSV-M. Normalized SARS-CoV-2 pseudovirus supernatants were gently mixed with 5 μg/mL biotin-HPA overnight at 4ºC. Pull-down samples attached on the streptavidin magnetic beads were washed 3 times in the NP-40 lysis buffer before boiling in the 2x Laemmli loading buffer. Protein samples were then separated by SDS-PAGE and detected by immunoblotting of full-length spike and S2.

Reviewer #3 (Remarks to the Author):
In “A novel interferon stimulated gene GALNT2 is required to restrict respiratory virus infections” Ran et al. set out to explore interferon stimulated genes in the pulmonary tissue of IFNAR KO mice infected with SARS-CoV2.
The authors identify several genes that are differentially downregulated in the WT and IFNAR KO background after SARS-CoV-2 infection. They then overexpress all 28 genes in Huh7 cells and quantified percentage of infected cells after SARS-CoV-2 infections. Here GALNT2 stands out as the one reducing infection the most.
Now focusing only on GALNT2, GALNT2 transcript levels are analyzed in single-cell RNA-seq data from BALF and PBMCs from healthy controls and COVID-19 patients – mild and severe cases. The authors state that high GALNT2 transcript level is associated with COVID-19 patients compared to Healthy Controls, but more with mild than severe disease.
Based on a number of cell-based in vitro assays and animal KO/overexpression of GALNT2 the authors conclude that GALNT2 is an interferon responsive gene with antiviral activity towards SRARS-CoV-2 as well as Influenza A virus.
The authors explore the impact of GALNT2 overexpression and conclude that there is no major impact on the host cell by measures of transcriptomics/proteomics and glycoproteomics and speculate that GALNT2functions through targeting viral proteins.
Using in vitro enzymatic glycosyltransferase activity assays, they then test the activity of GALNT2 on peptides from the spike protein covering a furin and a TMPRSS2 cleavage site. This data leads the authors to test if GALNT2 activity changes processing of recombinantly expressed spike and HA with similar cleavage sites. The authors conclude that both Spike and HA are differentially processed after GALNT2 activity and suggest that impaired viral entry could be part of the anti-viral mechanism.
In cell-cell fusion assays etc.
Comments
In general, the figures are laid out in a logic manner and the manuscript flow is relatively easy to follow. The manuscript text could benefit from some sort of language service/proofreading.
We are deeply grateful to the reviewer for your appreciation of our research. In accordance with the suggestion, we have carried out a comprehensive review and refinement of the manuscript.

Query 1. One argument the authors provide that GALNT2 is anti-viral relates to four alleged GALNT2 LoF variants identified in the GnomAD database. Prior studies exploring such variants present in the broadpopulation suggested none or only minor effect on the enzymatic activity (e.g. Guda et al., PNAS 2009, Hansen et al., Glycobiology 2015). Revisiting this database, this reviewer did not find any of the suggested variants to be predicted to be deleterious and two are even validated as benign. Maybethe authors tested the enzymatic activity of the 4 variants (V554M, V552M, D314A, Q216H)? If so, please provide this data to support the conclusion that GALNT2 activity is anti-viral.

We apologize for not paying enough attention to prior WES analysis on GALNTs and corresponding validation of enzyme actives. We previously used the term "loss-of-function" to describe genetic variants of GALNT2 that lost their ability to suppress SARS-CoV-2 viral activities. However, this does not necessarily correlate with a "loss-of-activity" of the corresponding GalNAc-T enzyme variants. The reviewer raised an important point regarding the enzymatic activities of these GALNT2 variants, which we have addressed in our subsequent experiments.

Indeed, three out of four missense mutations of GALNT2 in this manuscript (V554M, D314A, Q216H) was previously identified by Hansen et. al. in 2015 in a WES of 2000 Danes (PMID: 25267602), however, none of those mutations were found to affect the enzyme activity of GALNT2 in an in vitro glycosylation assay. Because our proposed mechanism of GALNT2 involved a peptide substrate derived from the TMPRSS2 cleavage site of SARS-CoV-2, which is different from mucin peptides commonly used in standard in vitro glycosylation assay, we re-tested the in vitro activity of these GALNT2 variants with the SARS-CoV-2 derived peptide. As demonstrated below, although there were no significant differences between enzymatic activities of those mutants and the WT GALNT2 in the overnight reactions, notable decline in enzymatic activities were observed with all four variants of GALNT2 when the reaction time was reduced to three hours (Figure B). This result thus indicated that the identified missense mutations of GALNT2 (V554M, V552M, D314A, Q216H) affected the kinetics of GALNT2 rather than completely inactivated the enzyme.

The previous studies by Hansen et. al. used a peptide derived from muc1 to test the enzyme activity of those GALNT2 variants, which has a much faster catalytic kinetics. This could have covered up the relatively minor impact of those missense mutations.

We included these data into the Extended Data Fig. 11c-d to strengthen our conclusions in the revised manuscript.

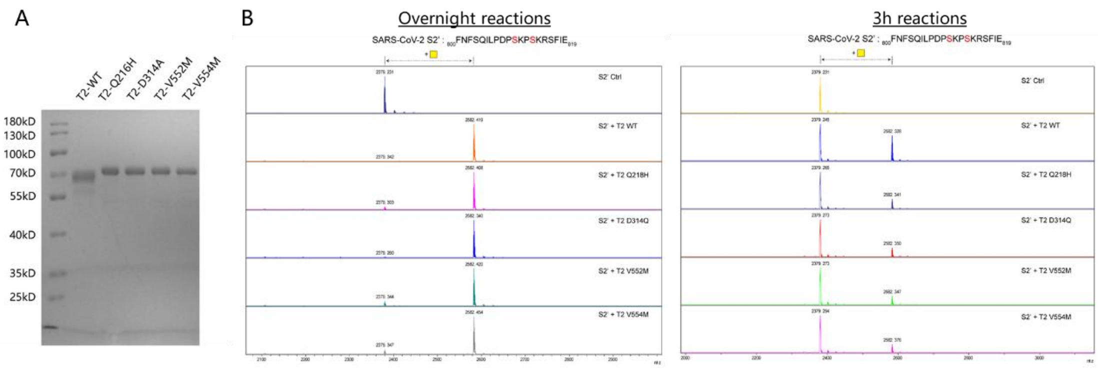


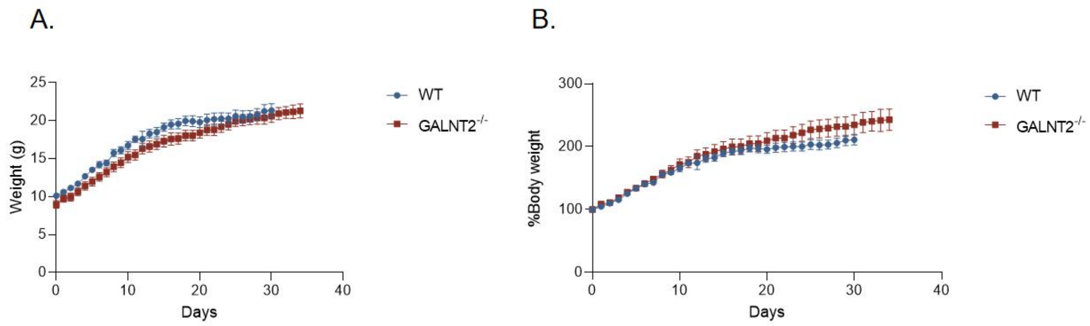
Figure legend: (A) Soluble GalNAc-T2 (WT and four mutations) were overexpressed in FreeStyleTM 293-F cells and purified by Ni-NTA chromatography. All proteins were analyzed by SDS-PAGE and Coomassie staining. (b-c) MALDI-TOF analysis of GalNAcylation reactions catalyzed by purified GalNAc-T2 on the synthetic spike S2' (TMPRSS2) peptides. Reactions were performed for overnight (B) or 3h (C) and analyzed as described in the

Methods. An increase of 203 Da corresponds to addition of one GalNAc residue was obvious detected O-glycosylation by WT GalNAc-T2, but an apparent decline in four mutants.

Query 2. Several GALNT2 LoF animal models have been generated and published, and significantly reduced body weight as well as a short stature phenotype have consistently been reported in Galnt2 deficient mice, as well as in humans and dairy cattle (Edmondson et al., BioRxiv 2024, Mercanoglu et al., Sci Rep 2024, Yang et al., PNAS 2023, Verzijl et al., Mol Metab 2022, Zilmer et al., Brain 2020, Khetarpal et al., Cell Metab 2016, Reynolds et al., Nat Genet 2021). Thus, the data presented in Fig. 3 awhere WT and Galnt2 KO animals have same body weight is a bit confusing. And this reviewer wonders if the reduced body weight observed post infection is an inherent phenotype of the animal. If the authors have generated a LoF model that is different from previously published, they should note this and present growth data. In general, the authors should discuss their findings in relation to the established literature.

We are very grateful for the valuable suggestions of the reviewer.

After the WT and GALNT2$^{-/-}$ mice were born, genotype identification was carried out within 3 weeks, including sequencing and the identification of protein expression levels (Extended Data Fig. 7a-c). After 3 weeks old of age, they were weaned and divided into litters, then weighed daily from 3 weeks to 8 weeks of age and compare their weights with those of WT mice. According to the body weight results (as shown below), from birth to 6 weeks of age, KO mice exhibited a slightly lower body weight compared to wild-type mice. However, after 6 to 8 weeks of age, the body weight of KO mice gradually increased and became comparable to that of wild-type mice. Overall, these findings suggest that GALNT2 KO may influence body weight in the early developmental stages, which aligns with previous studies. It should be noted that our mouse infection experiments were performed using mice aged 6 to 8 weeks.



Mice age: From 3 weeks to 8 weeks

Figure legend: Uninfected wild-type (WT) and GALNT2 KO mice undergo genetic identification and weaning within three weeks after birth. Starting from the third week post-weaning, the daily body weight of the mice was recorded (A).

In mice infection experiments, we defined the body weight of mice on the day of infection as 100% and subsequently weighed mice daily. Since GALNT2 KO does not cause significant body weight reduction after 6 weeks of age, the observed reduction

in body weight post-infection represents a phenotypic response attributable to viral infection.
As suggested, we discussed our GALNT2$^{-/-}$ mouse model in more detail in the revised manuscript (lines 248-250).

Query 3. To explore the impact of Galnt2 overexpression the authors take a multi-omics approach with transcript omics/proteomics/glycoproteomics and identify significant changes at all three levels, but do not discuss or add value to these changes. Interestingly Galnt2 overexpression in mouse pancreas is lethal at the age of 4-6 weeks and other studies have demonstrated dose-dependent activities of GALNT2 (Mercanoglu et al., Sci Rep 2024, Hintze et al., JBC 2019). Of note, several of the upregulatedproteomics/glycoprotomics hits are ER residents, which could be suggestive of a stressed secretory pathway, as often seen when driving high protein expression. Maybe there are technical details worth exploring in this context, but this reviewer failed to find the experimental section describing transcriptomics /proteomics/ glycoproteomics experiments? How were these experiments conducted?
As suggested, we have provided detailed experimental protocols for the RNA-seq, proteomics, and glycoproteomics analyses conducted both in the presence and absence of GALNT2 in the revised manuscript (lines 1019-1067). The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD051101. The accession number for the Next-gen RNA sequence data reported in this paper is Gene Expression Omnibus (GEO) database: GSE301783.

Based on our omics results, we did not observe such obvious indirect effects, which further confirms that *O*-linked glycosylation of the spike protein is the primary target of GALNT2 during viral infections (lines 295-305, 523-533). Precisely due to these multi-omics findings, we concentrated our research on the targets of GALNT2 on viral proteins. Our experimental data (Fig 4 and 5) also demonstrates that GALNT2 directly targets viral glycoproteins as its main mechanism for antiviral activity.

Interestingly, we reanalyzed the glycosylation omics data (as shown below) and indeed observed an enrichment of GALNT2-modified substrates in the endoplasmic reticulum, which aligns with the known cellular localization of GALNT2 (previously identified by Hansen et. al. in 2019). However, these substrates are not ISGs and have not been previously associated with viral infections. Therefore, we believe this data does not directly relate to the central topic of our study, and we do not intend to include it in the manuscript. We will continue to study their impact on viral infection in future.

In in vivo experiment, we delivered GALNT2 based on AAV9 vector intranasally, therefore GALNT2 is primarily expressed in the lung, not the pancreas, and mice did not show any obvious clinical symptoms and weight changes before virus infection.

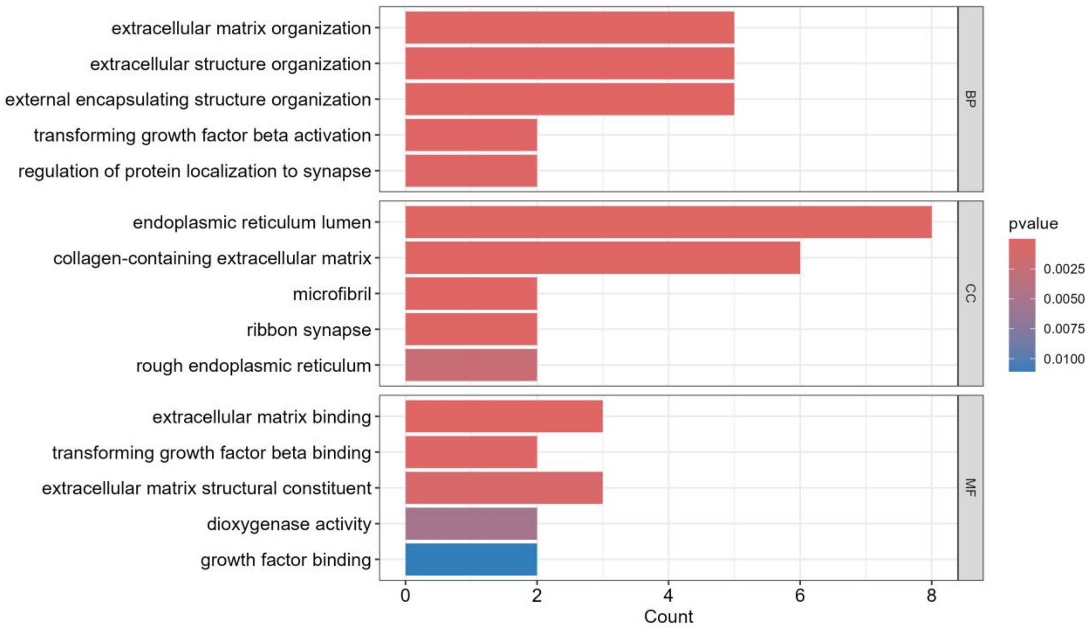


Figure legend: GO enrichment of proteins with upregulated O-glycosylation in GALNT2-OE A549 cells compared with control cells. Greater than 1.5-fold increase in abundance upon GALNT2 overexpression and lower than 0.2 adjusted p value were set as the threshold criteria for GalNAc-T2-regulated proteins.

Query 4. GALNT2 is part of a large family of isoenzymes with largely redundant activities. To substantiate their idea that GALNT2 specifically has anti-viral activity, the authors express several GALNT isoenzymes to test in vitro activity with spike and HA peptides. The authors do not present data to suggest that the other isoforms are active but write that a peptide from muc1 is tested against all isoforms demonstrating enzymatic activity. The authors should provide this data as at least one of the isoforms should not work on this peptide? (Coelho et al., JACS 2022)

We thank the reviewer for highlighting the omission of positive control data for the in vitro analysis of GalNAc-T isozymes. We do have these data, as described in the Methods section (lines 993-995), as shown below, all isozymes except GalNAc-T7 and T18 demonstrated activity towards the positive control peptide derived from MUC1. GalNAc-T7 is known to modify pre-glycosylated substrates (PMID: 25155433) and has been utilized in our previous study (PMID: 38755139). GalNAc-T18 belongs to the Y-family of GalNAc-Ts and does not have a known substrate (PMID: 22171061, 30797803). Therefore, we consider our positive control data to be appropriate.

Since we published these data before (PMID: 38755139), have chosen not to republish the data and to cite the aforementioned references (reference 59) to explain why we believe GalNAc-T7 is active despite showing no obvious activity towards the MUC1 peptide in the revised manuscript.

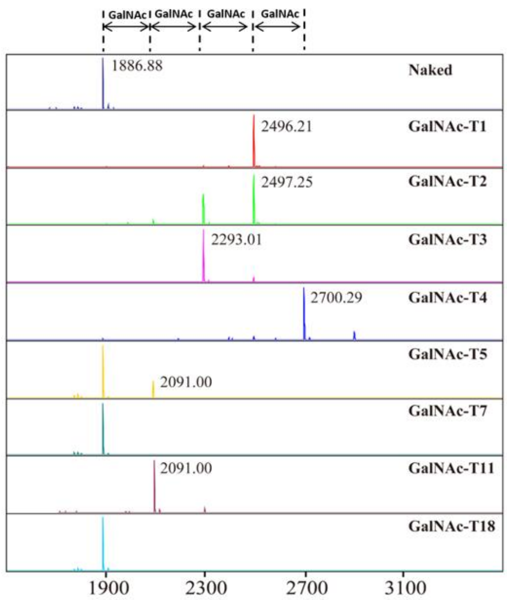


Figure legend: MALDI-TOF analysis of GalNAcylation reactions catalyzed by purified GalNAc-Ts on a synthetic Muc1-derived peptide (HGVTSAPDTRPAP GSTAPPA). Reactions were performed and analyzed as described in the Methods. An increase of 203 Da corresponds to the addition of one GalNAc residue to the peptide.

Reviewers Comments:

Reviewer #1 (Remarks to the Author):

The authors have addressed my concerns and I congratulate them to an interesting study.

We are very grateful to this reviewer for his/her very positive and professional comments, which have significantly improved the quality of our manuscript.

Reviewer #2 (Remarks to the Author):

I want to thank the authors for thoroughly addressing my points.

This is an excellent manuscript, a pleasure to read - I fully support its publication.

We are very grateful to this reviewer for his/her very positive and professional comments, which have significantly improved the quality of our manuscript.

Note: Fig 4I: Hoescht33342 should be Hoechst33342

Changed.

Reviewer #3 (Remarks to the Author):

We appreciate the time and effort the reviewer dedicated to providing the valuable and professional suggestions for improving our work.

For Query#1

The MALDI-TOF time course experiment is not quantitative, and the conclusion written in the manuscript text (Line 334-338) is exaggerated and incorrect. Based on this experiment the authors cannot conclude that the activity of GALNT2 is markedly decreased in the missense variants and should adjust their conclusions accordingly.

We fully agree that MALDI-TOF, a classical way to measure the enzyme activity of *O*-GalNAc transferases, has limited quantitation ability. We therefore follow the reviewer's suggestion to tone down our conclusion in the main text (line 334-338) so that it reads as below (line 242-250 in revised manuscript):

"Next, we checked whether the inability of the previously identified missense mutations of GALNT2 (V554M, V552M, D314A, Q216H) to inhibit SARS-CoV-2 infection was due to impaired enzymatic activity of GALNT2 towards S810/813 of the spike protein using the *in vitro* glycosylation assay. Interestingly, we did observe a noticeable decrease of glycosylated peptide with all four variants of GALNT2 when the reaction time was reduced to three hours (Extended Data Fig. 11d). However, due to the limited quantitation capability of MALDI-TOF-based assays, we could not rule out the possibility of those missense mutations affecting GALNT2 through other mechanisms, such as transcriptional regulation. Notably, three out of four missense mutations of GALNT2 (V554M, D314A, Q216H) were found to have no effect on the enzyme activity of GALNT2 when tested against a mucin-based peptide substrate (PMID: 25267602)."

Line 213 - Variants found in the GnomAD NOT GenomeAD

Changed.

The authors use of LOF in relation to an enzyme (here GALNT2) should be defined as conventionally this refers to loss of activity and is misleading in the context presented in the abstract (line 60) and in the manuscript (line 212.

We replaced LOF (loss-of-function) with loss of antiviral function in this manuscript (line 61-62 in revised manuscript).

Query#2

No further comments

Query#3

GALNT2 is NOT localized in the ER, but in the Golgi.

And the conclusion line 301 seems exaggerated and incorrect. The findings of ER proteins should be mentioned and may be a consequence of cellular stress.

We fully agree with this reviewer that the impact of GALNT2 overexpression on ER proteins (Extended Data Fig. 9c) should be mentioned in main text. We therefore follow the reviewer's suggestion and modified the text from line 298 to line 303 as below (line 212-219 in revised manuscript):

"As depicted in Extended Data Fig. 9a-b, our findings demonstrated that GALNT2 overexpression exerts minimal influence on the expression of a limited subset of host genes at both the RNA and protein levels. As an important member of the *O*-GalNAc transferase family, the overexpression of GALNT2 did increase the *O*-glycosylation levels of a selected group of substrate proteins, including a few proteins in the endoplasmic reticulum (Extended Data Fig. 9c). As GALNT2 is a Golgi-localized enzyme, this might be a consequence of cellular stress caused by overexpression, but none of those have been reported with viral infections."

Query#4

No further comments